

Electromagnetic Induction

CHAPTER SIX
CLASS - XII

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Learning Objectives:

- Electromagnetic induction
- Faraday's laws
- Induced EMF and current
- Lenz's Law
- Motional emf
- Self and mutual induction.
- AC generator

Introduction

- Electricity and magnetism were once thought unrelated.
- Oersted and Ampère showed currents produce magnetic fields.
- Faraday and Henry (1830) discovered:
They demonstrated that **changing magnetic fields** can **induce electric currents** in a closed coil — this is called **electromagnetic induction**.
- Enabled inventions like **generators** and **transformers**.
- Faraday's Remark:- When asked about the use of his discovery, he replied:

THE EXPERIMENTS OF FARADAY AND HENRY

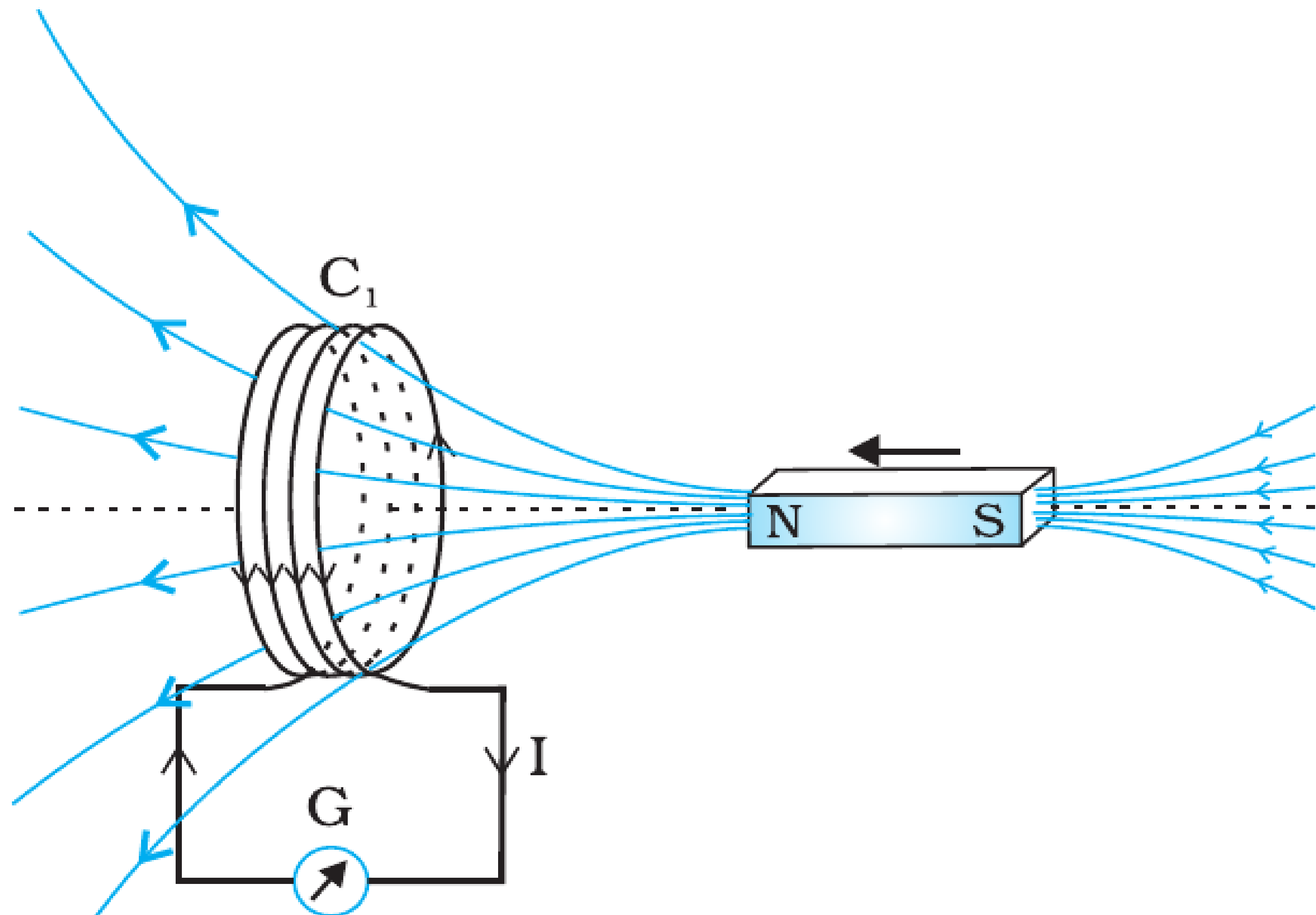


FIGURE 6.1 When the bar magnet is pushed towards the coil, the pointer in the galvanometer G deflects.

Experiment 1: Magnet and Coil

- A bar magnet moves relative to a coil connected to a galvanometer.
- Current is induced only during motion.
- Faster motion $\rightarrow$ greater current (larger deflection).
- Direction of induced current reverses with motion.
- Key insight: **Relative motion** causes induction.

THE EXPERIMENTS OF FARADAY AND HENRY

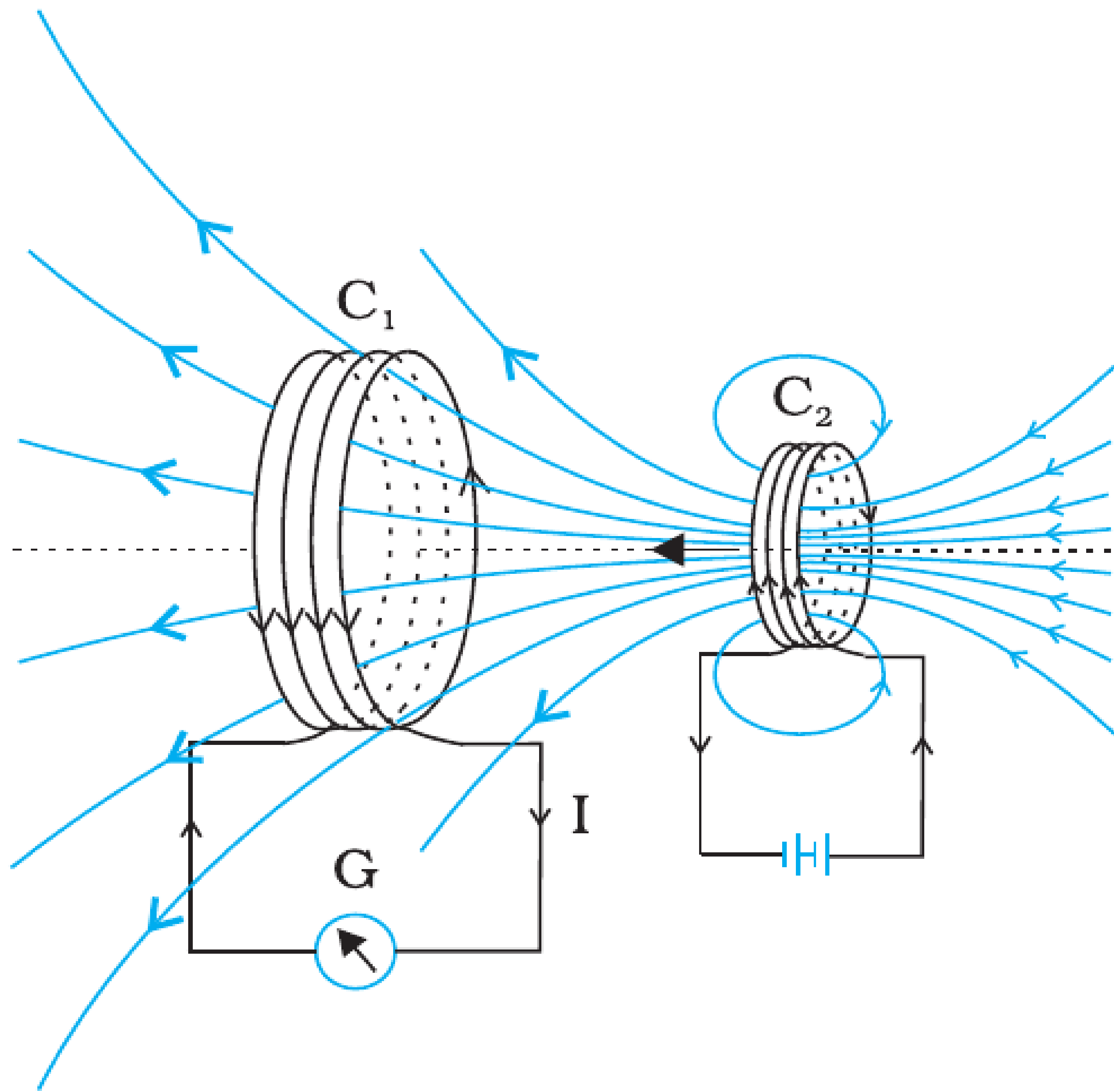


FIGURE 6.2 Current is

Experiment 2: Two Coils

- Coil C_2 carries a current and is moved near coil C_1 (with galvanometer).
- Induced current is observed in C_1 when:
 - C_2 is moved, or
 - C_1 is moved relative to C_2 .
- Again, **relative motion** is essential.

THE EXPERIMENTS OF FARADAY AND HENRY

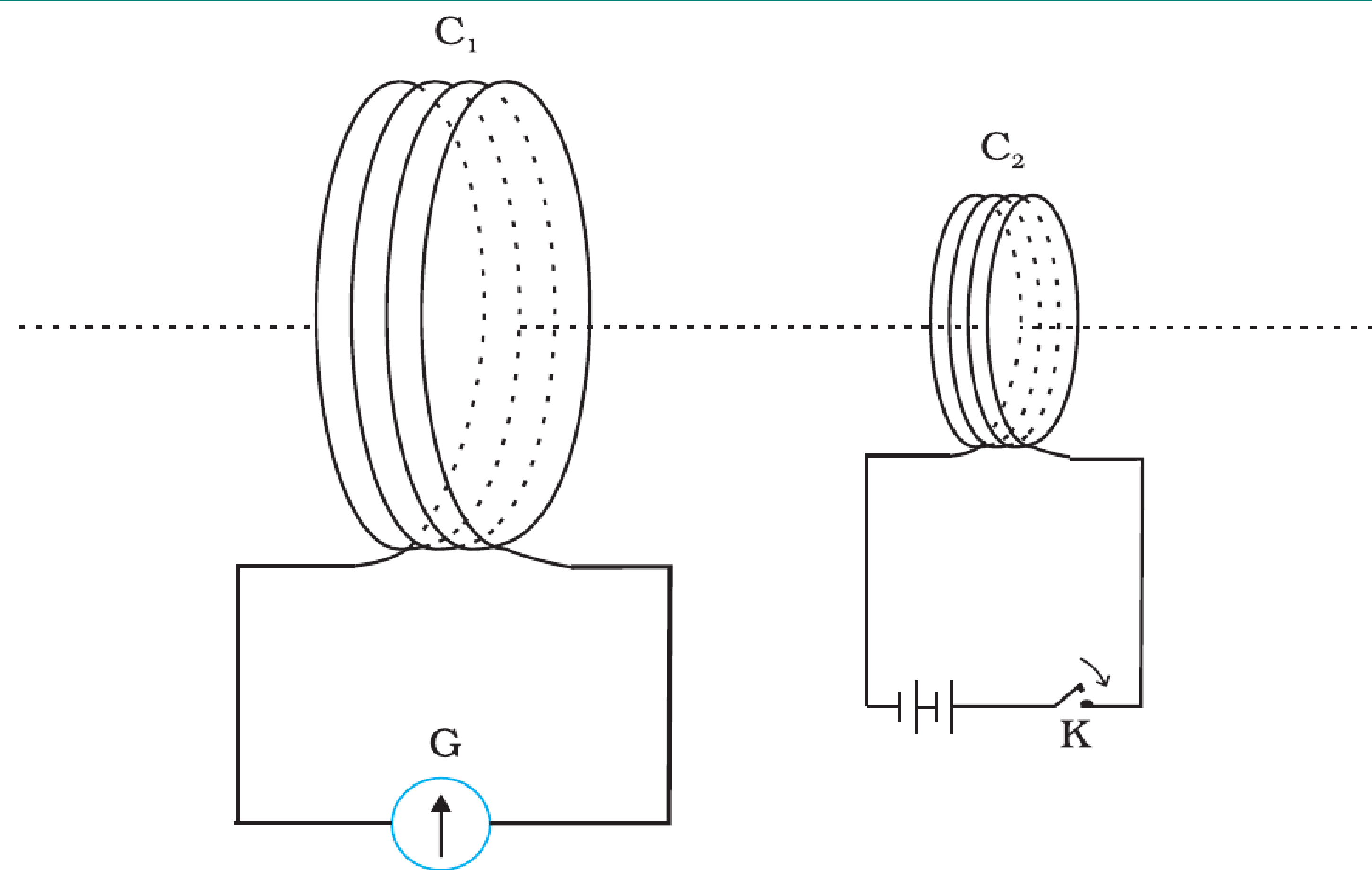


FIGURE 6.3 Experimental set-up for Experiment 6.3.

Experiment 3: Stationary Coils

- Both coils are stationary.
- Coil C_2 is connected to a battery via a tapping key.
- **Momentary current is observed in C_1 , when The key is pressed or released.**
- **No motion needed — only a changing magnetic field.**

Faraday's Law of Induction

- Magnetic Flux: $\Phi_B = BA \cos \theta$
- Faraday's Law: The **induced emf (electromotive force)** in a coil is **equal to the rate of change of magnetic flux** through the coil.

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

- For N turns:

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

- Negative sign $\rightarrow$ **Lenz's Law**: Induced emf opposes change in flux.
- Ways to vary flux: Change B , area A , or angle θ

Example 6.1 Consider Experiment 6.2. (a) What would you do to obtain a large deflection of the galvanometer? (b) How would you demonstrate the presence of an induced current in the absence of a galvanometer?

Solution

- (a) To obtain a large deflection, one or more of the following steps can be taken: (i) Use a rod made of soft iron inside the coil C_2 , (ii) Connect the coil to a powerful battery, and (iii) Move the arrangement rapidly towards the test coil C_1 .
- (b) Replace the galvanometer by a small bulb, the kind one finds in a small torch light. The relative motion between the two coils will cause the bulb to glow and thus demonstrate the presence of an induced current.

In experimental physics one must learn to innovate. Michael Faraday who is ranked as one of the best experimentalists ever, was legendary for his innovative skills.

EXAMPLE 6.2

Example 6.2 A square loop of side 10 cm and resistance $0.5\ \Omega$ is placed vertically in the east-west plane. A uniform magnetic field of 0.10 T is set up across the plane in the north-east direction. The magnetic field is decreased to zero in 0.70 s at a steady rate. Determine the magnitudes of induced emf and current during this time-interval.

Solution The angle θ made by the area vector of the coil with the magnetic field is 45° . From Eq. (6.1), the initial magnetic flux is

$$\begin{aligned}\Phi &= BA \cos \theta \\ &= \frac{0.1 \times 10^{-2}}{\sqrt{2}} \text{ Wb}\end{aligned}$$

Final flux, $\Phi_{\min} = 0$

The change in flux is brought about in 0.70 s. From Eq. (6.3), the magnitude of the induced emf is given by

$$\varepsilon = \frac{|\Delta \Phi_B|}{\Delta t} = \frac{|(\Phi - 0)|}{\Delta t} = \frac{10^{-3}}{\sqrt{2} \times 0.7} = 1.0 \text{ mV}$$

And the magnitude of the current is

$$I = \frac{\varepsilon}{R} = \frac{10^{-3} \text{ V}}{0.5 \Omega} = 2 \text{ mA}$$

Note that the earth's magnetic field also produces a flux through the loop. But it is a steady field (which does not change within the time span of the experiment) and hence does not induce any emf.

EXAMPLE 6.2

Example 6.3

A circular coil of radius 10 cm, 500 turns and resistance $2\ \Omega$ is placed with its plane perpendicular to the horizontal component of the earth's magnetic field. It is rotated about its vertical diameter through 180° in 0.25 s. Estimate the magnitudes of the emf and current induced in the coil. Horizontal component of the earth's magnetic field at the place is $3.0 \times 10^{-5}\ \text{T}$.

Solution

Initial flux through the coil,

$$\begin{aligned}\Phi_{B \text{ (initial)}} &= BA \cos \theta \\ &= 3.0 \times 10^{-5} \times (\pi \times 10^{-2}) \times \cos 0^\circ \\ &= 3\pi \times 10^{-7} \text{ Wb}\end{aligned}$$

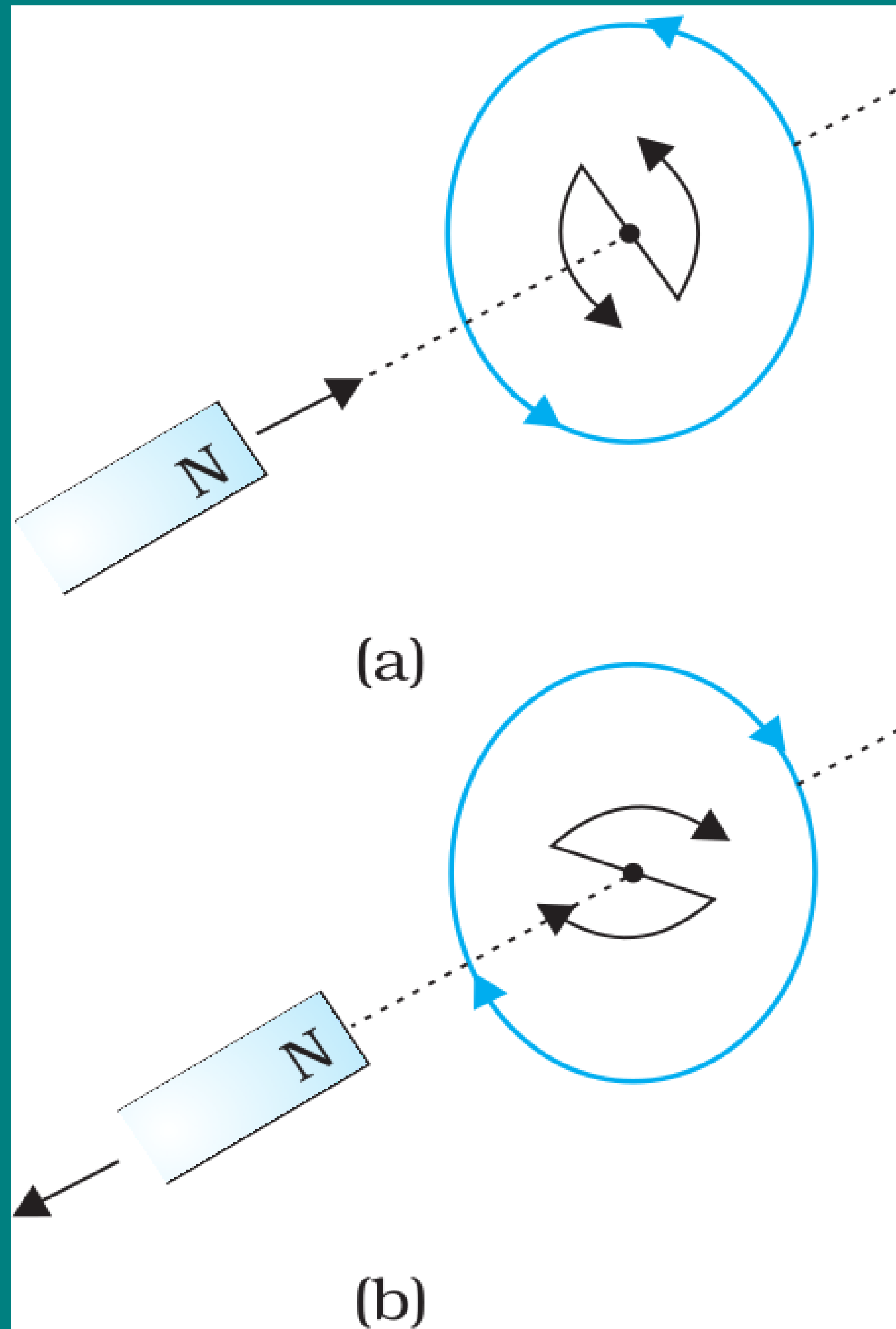
Final flux after the rotation,

$$\begin{aligned}\Phi_{B \text{ (final)}} &= 3.0 \times 10^{-5} \times (\pi \times 10^{-2}) \times \cos 180^\circ \\ &= -3\pi \times 10^{-7} \text{ Wb}\end{aligned}$$

Therefore, estimated value of the induced emf is,

$$\begin{aligned}\varepsilon &= N \frac{\Delta \Phi}{\Delta t} \\ &= 500 \times (6\pi \times 10^{-7}) / 0.25 \\ &= 3.8 \times 10^{-3} \text{ V}\end{aligned}$$

Lenz's Law



- **Lenz's law** which gives the **polarity (direction)** of the **induced emf** in a clear and concise fashion. **The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produced it.**
- As the **North-pole** of the bar magnet **moves towards the coil**, the **magnetic flux** through the coil **increases**.

- **Conservation of Energy:**

- Lenz's Law ensures the induced current does **not violate the law of conservation of energy**.
- To maintain motion (e.g., moving a magnet), **external work must be done**.
- This mechanical energy is converted into:
- **Electrical energy** (induced current), Then into **heat energy**.

- **Why Lenz's Law is Necessary:**

- If the induced current aided the change in flux:
It would result in continuous acceleration without input — a **perpetual motion machine**. This violates the **law of conservation of energy**.
- Lenz's Law prevents this by ensuring **energy balance**.

- **Quick Example:**

- Moving a bar magnet towards a loop → loop resists by creating opposing magnetic field → motion slows → **you must apply force** → energy is converted into **electric current** → dissipated as **heat**.

Example 6.4

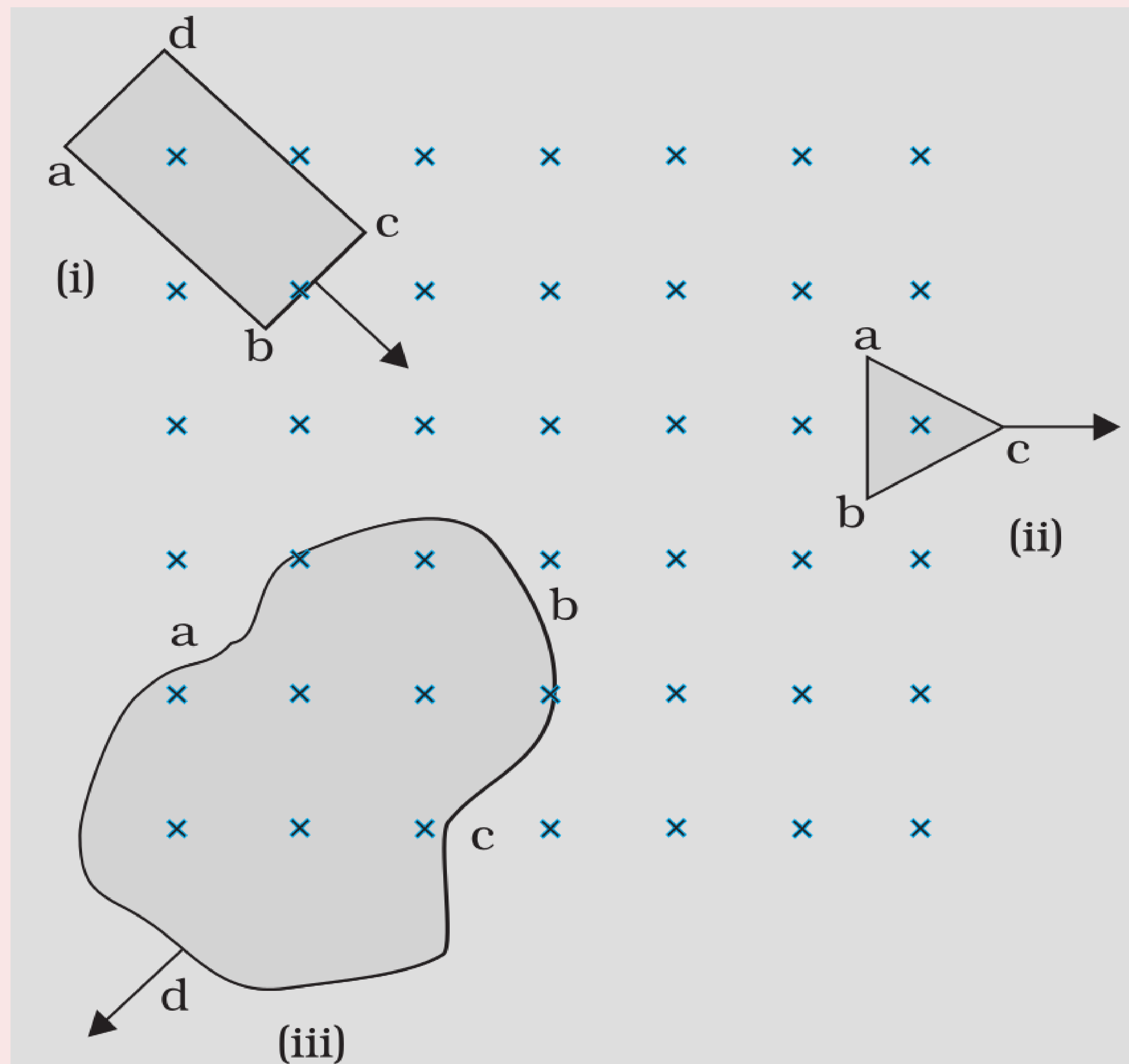


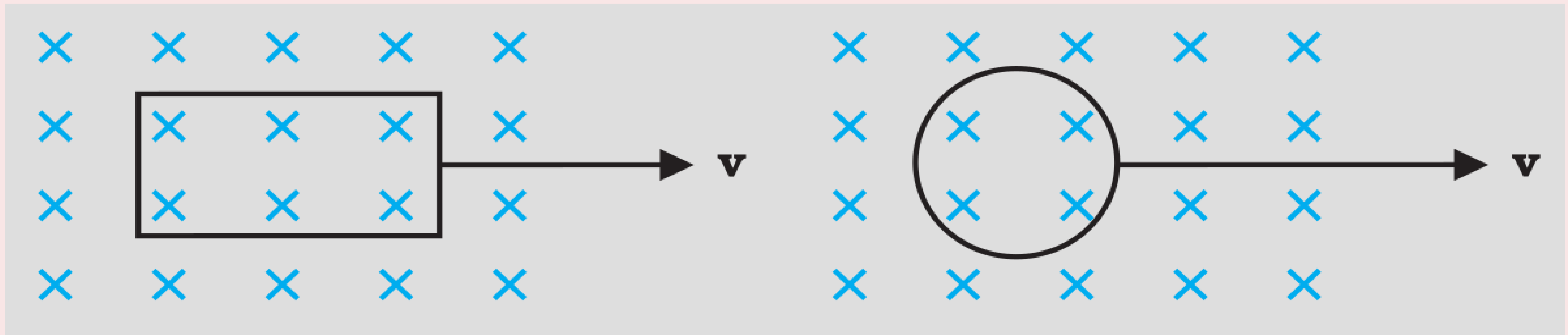
Figure 6.7 shows planar loops of different shapes moving out of or into a region of a magnetic field which is directed normal to the plane of the loop away from the reader. Determine the direction of induced current in each loop using Lenz's law.

Example 6.5

- (a) A closed loop is held stationary in the magnetic field between the north and south poles of two permanent magnets held fixed. Can we hope to generate current in the loop by using very strong magnets?
- (b) A closed loop moves normal to the constant electric field between the plates of a large capacitor. Is a current induced in the loop
 - (i) when it is wholly inside the region between the capacitor plates
 - (ii) when it is partially outside the plates of the capacitor? The electric field is normal to the plane of the loop.

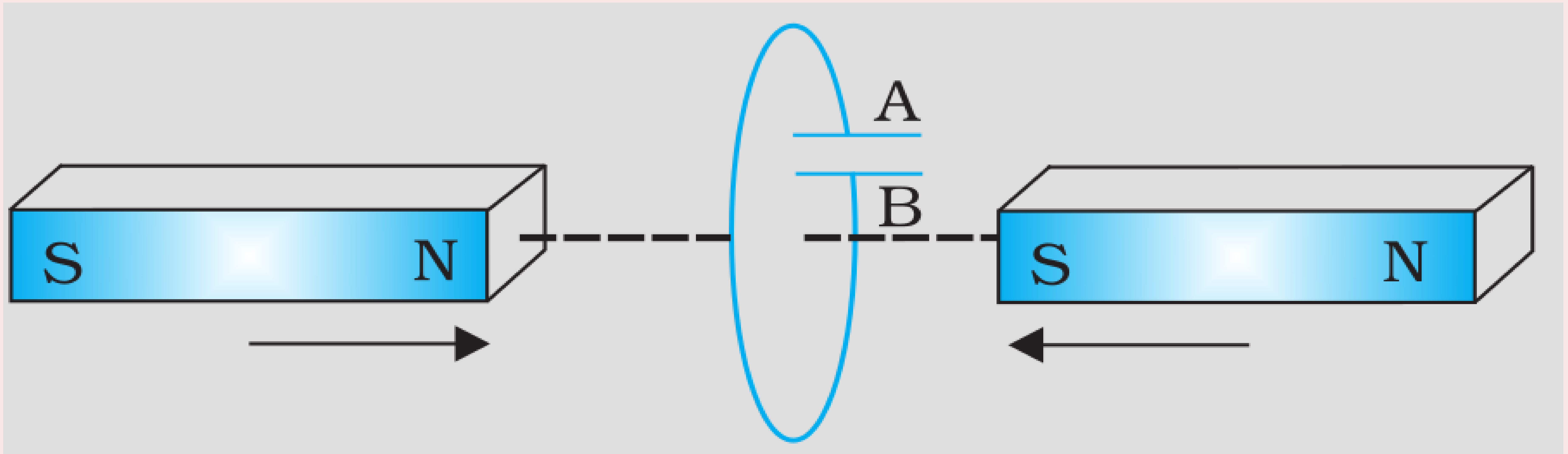
Example 6.5

- (c) A rectangular loop and a circular loop are moving out of a uniform magnetic field region (Fig. 6.8) to a field-free region with a constant velocity $\mathbf{v}$. In which loop do you expect the induced emf to be constant during the passage out of the field region? The field is normal to the loops.



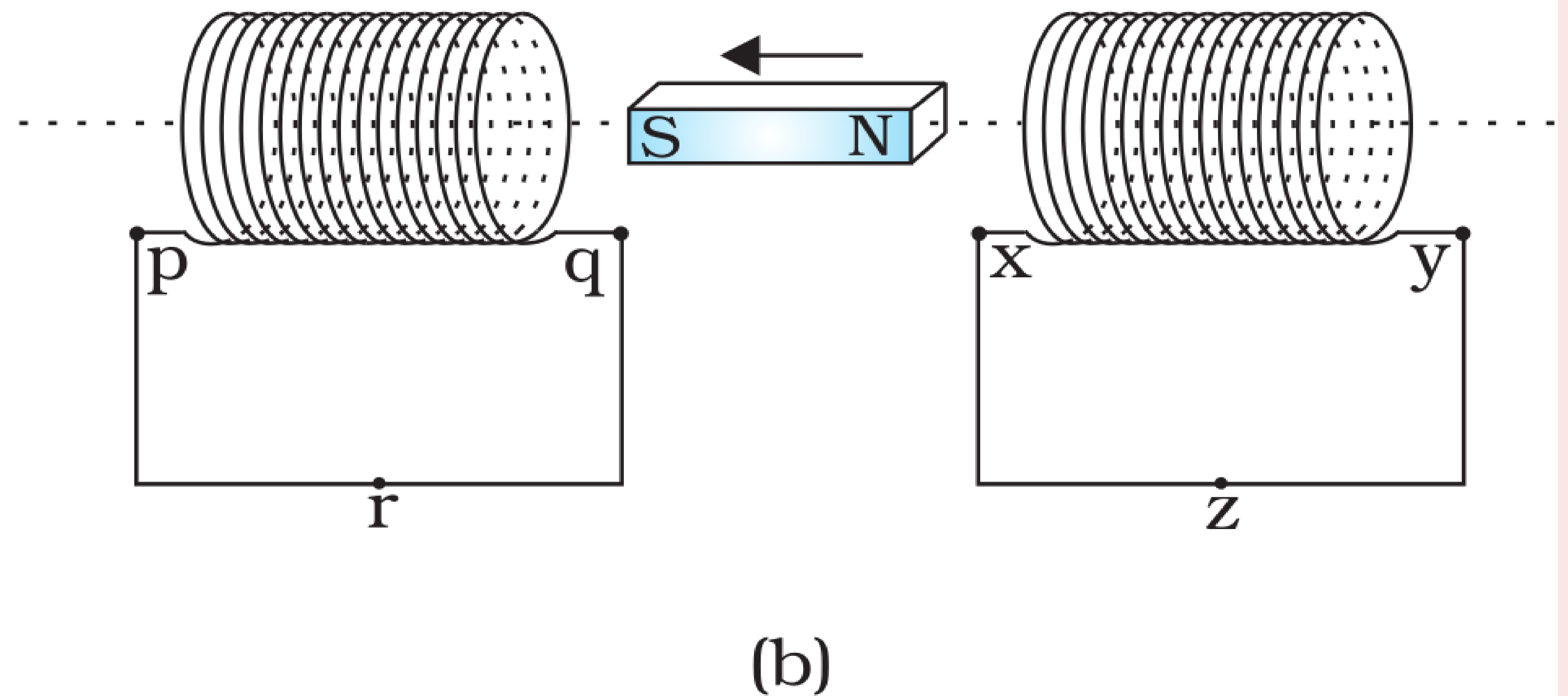
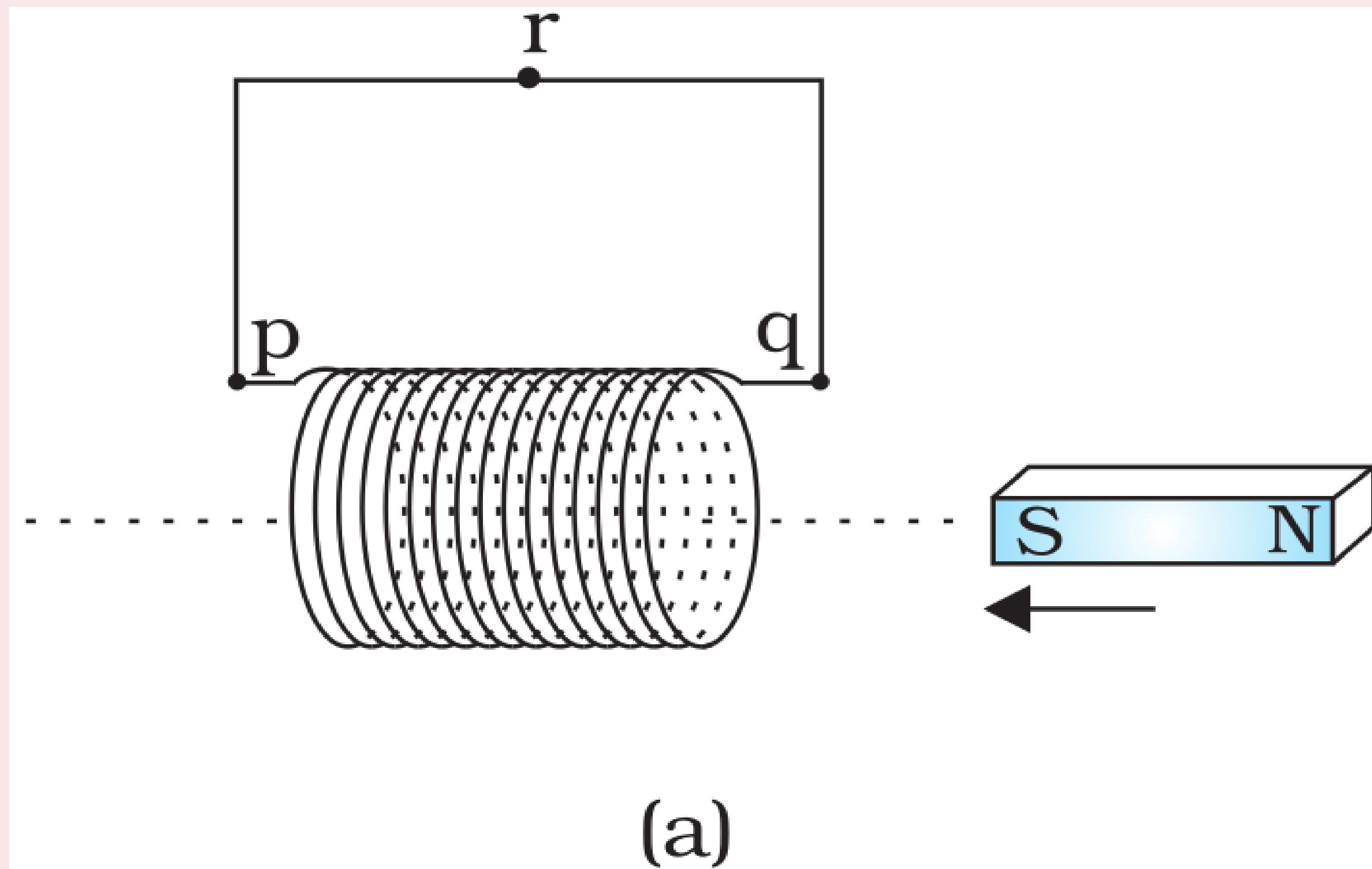
Example 6.5

(d) Predict the polarity of the capacitor in the situation described by Fig. 6.9.

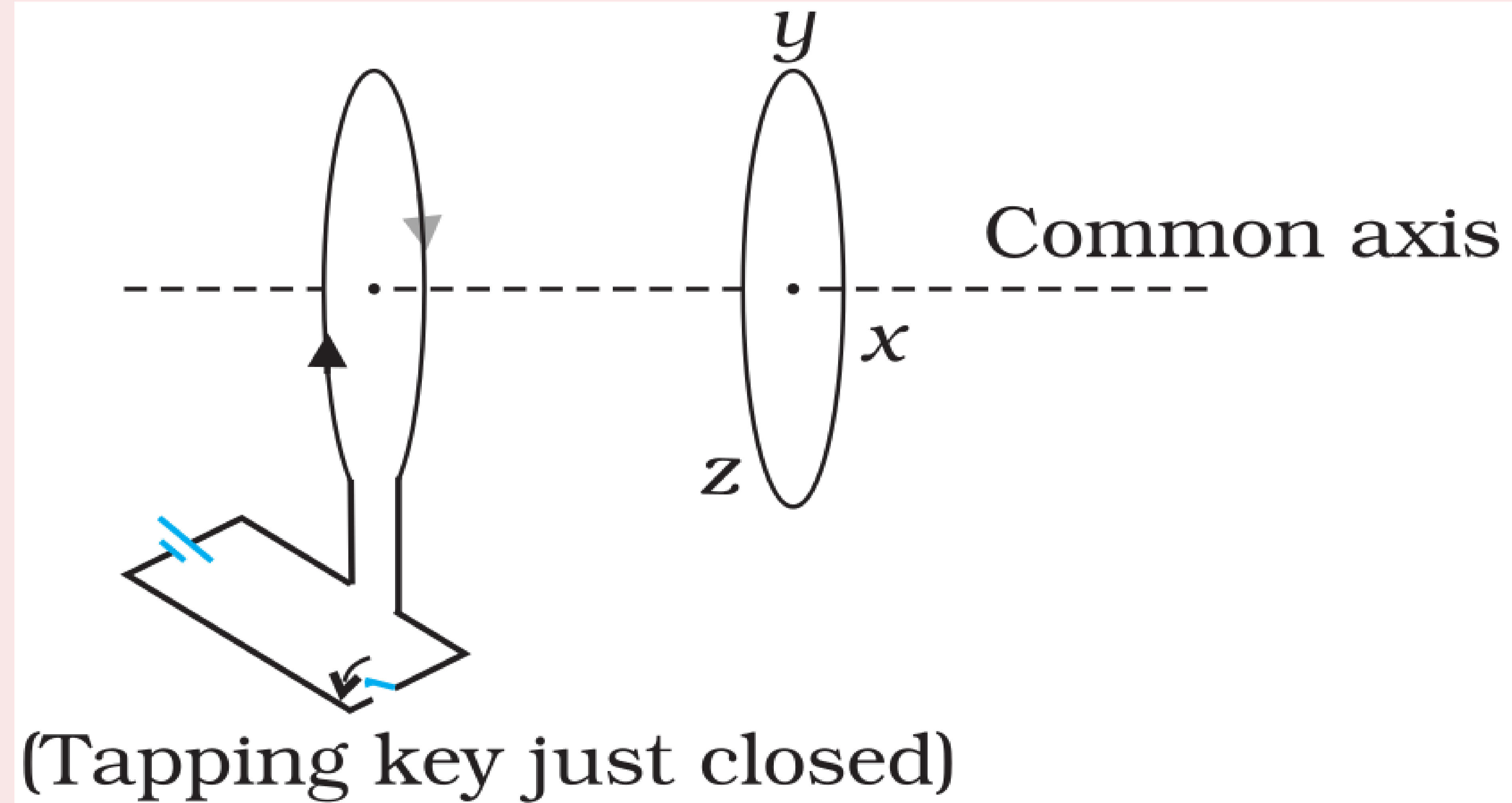


Exercise 6.1

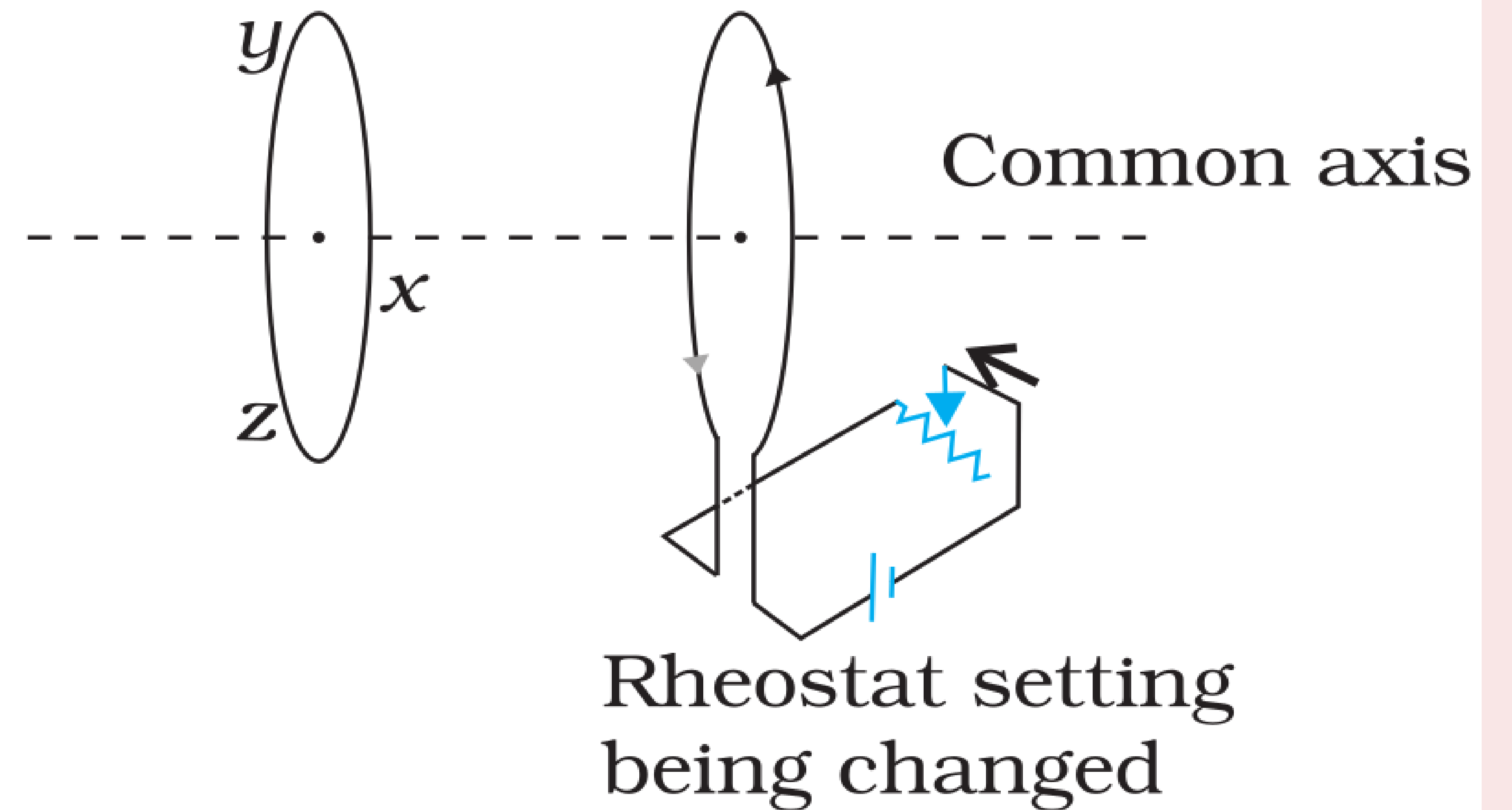
Predict the direction of induced current in the situations described by the following Figs. 6.18(a) to (f).



Exercise 6.1

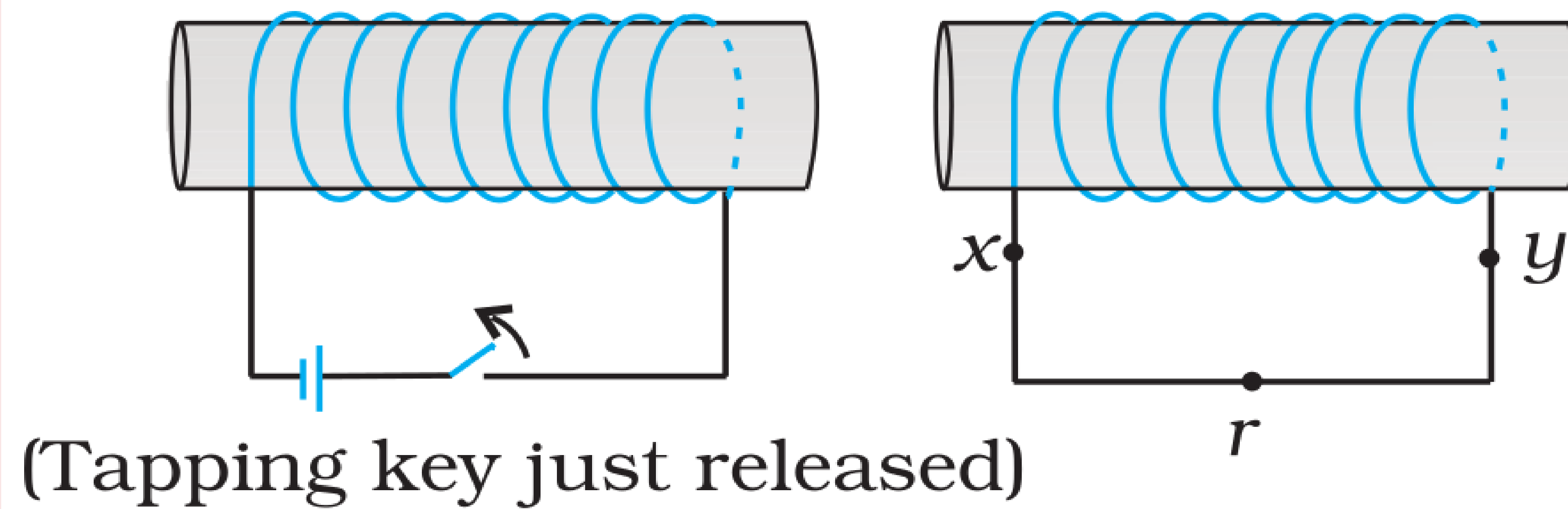


(c)

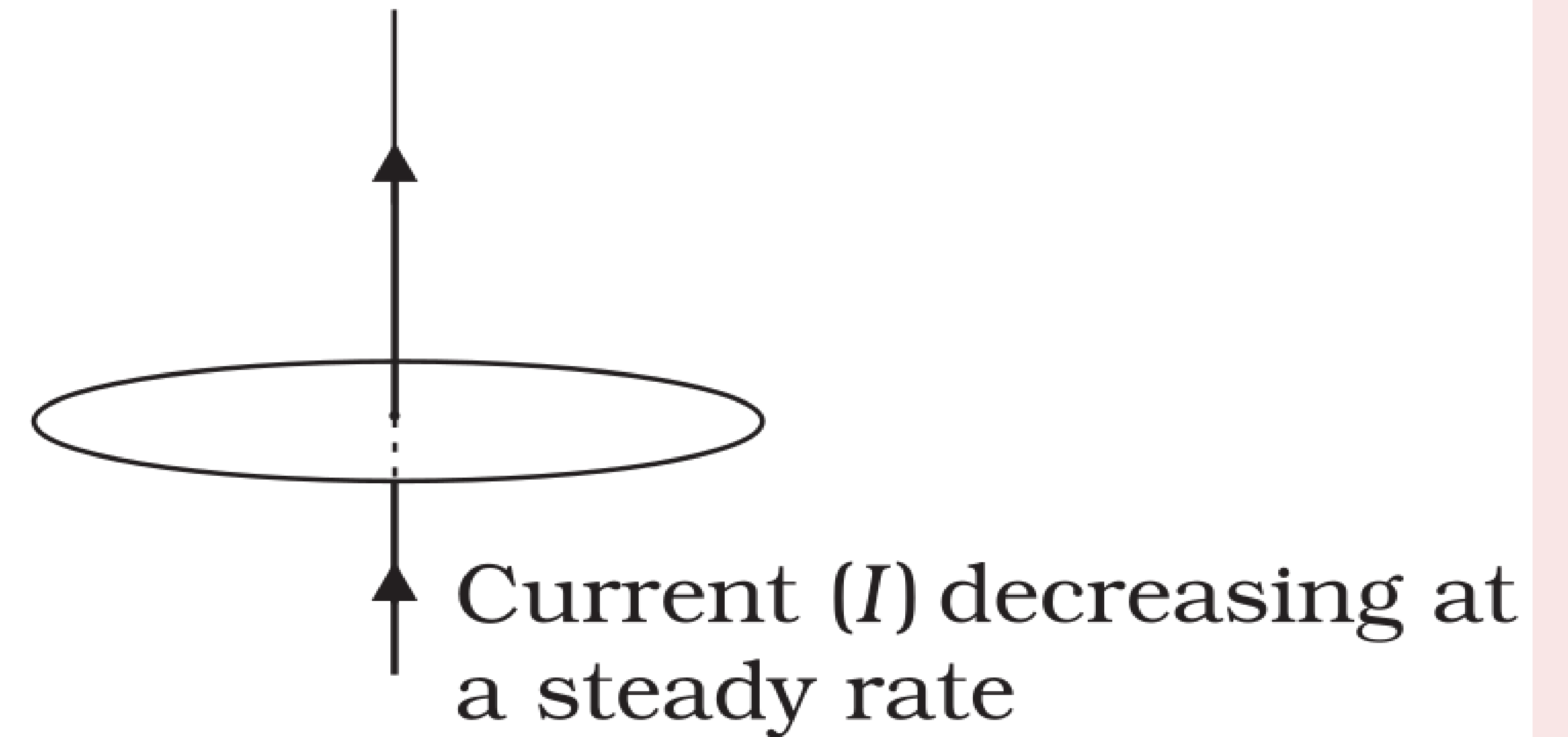


(d)

Exercise 6.1



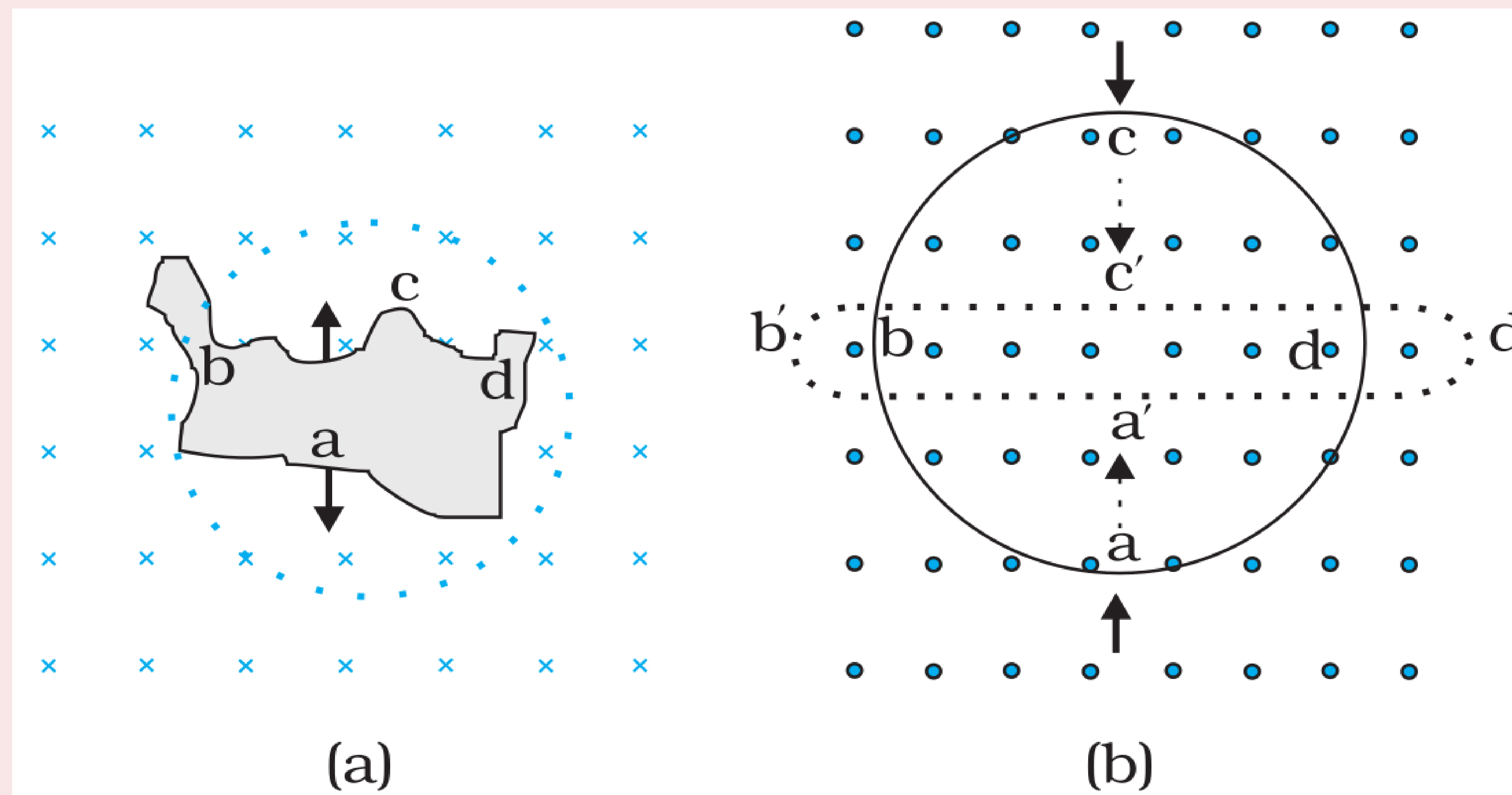
(e)



(f)

Exercise 6.2

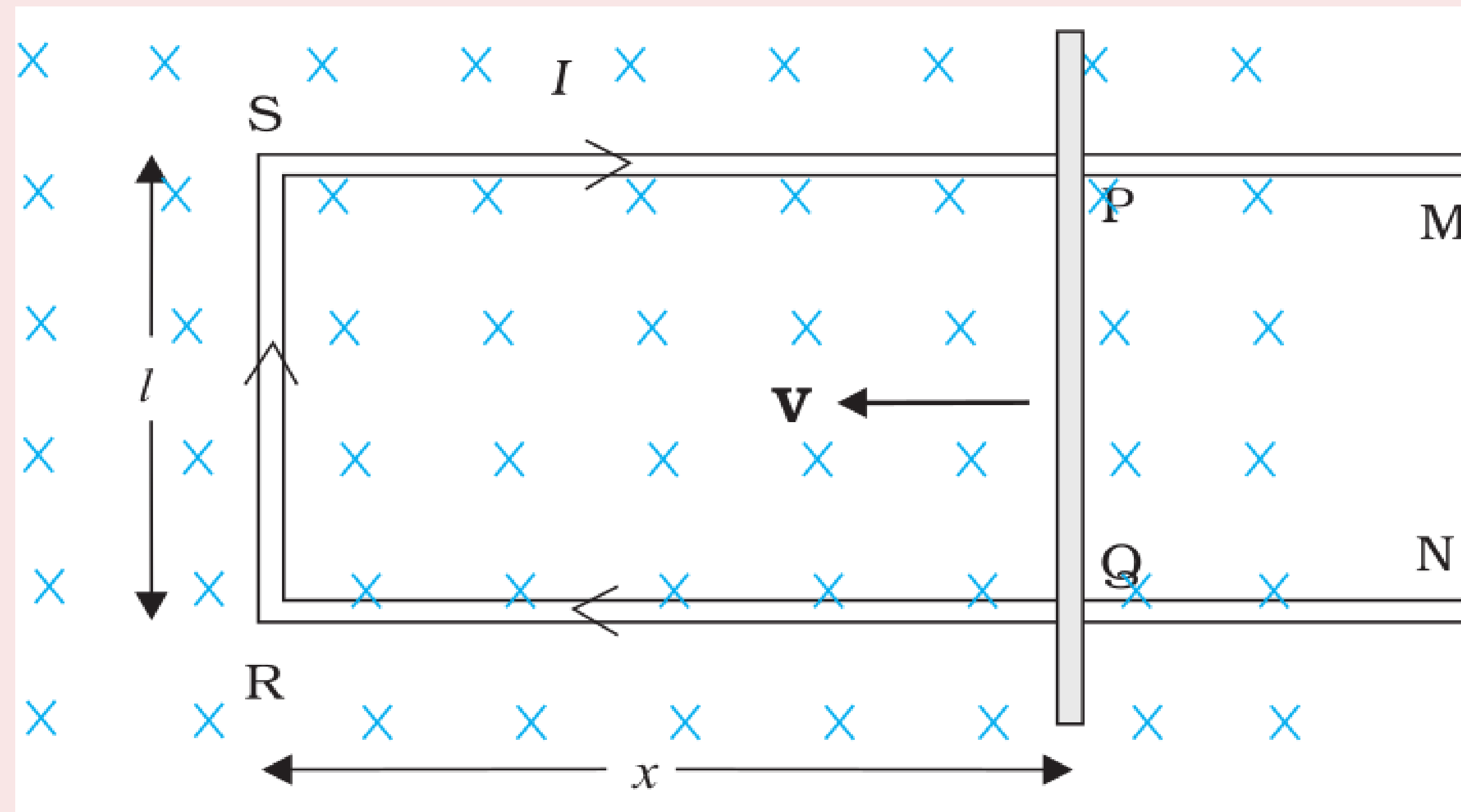
Use Lenz's law to determine the direction of induced current in the situations described by Fig. 6.19: **(a)** A wire of irregular shape turning into a circular shape; **(b)** A circular loop being deformed into a narrow straight wire.



Exercise 6.3

A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1 s, what is the induced emf in the loop while the current is changing?

Motional Electromotive Force



- A **conducting rod PQ** of **length l** is **moving** on a rail with **velocity v** in the **uniform magnetic field B** perpendicular to the plane of motion.

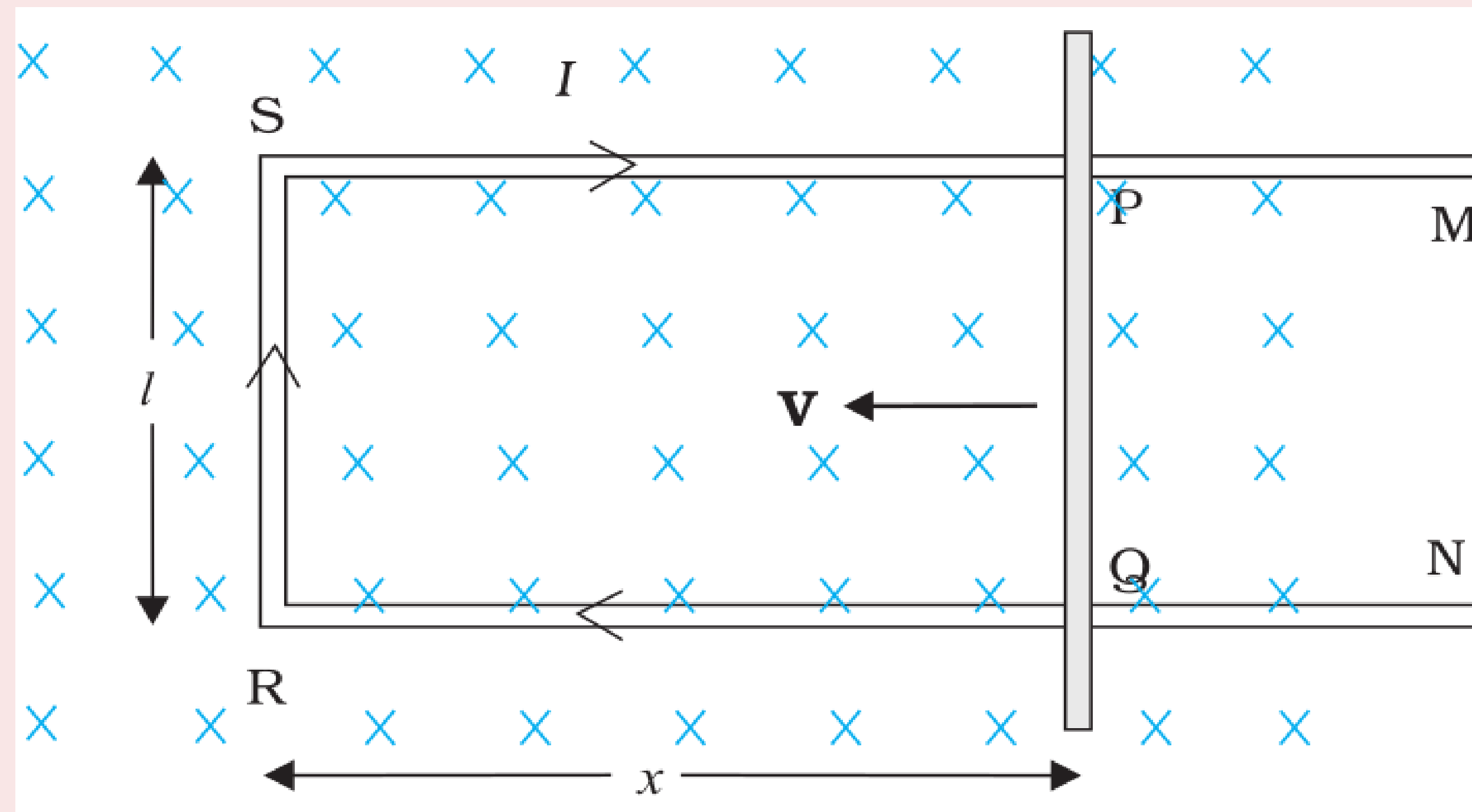
- If the length $RQ = x$ and $RS = l$, the **magnetic flux** ϕ_B enclosed by the **loop PQRS** will be

$$\phi_B = B \cdot A = Blx$$

- Since **x** is **changing with time**, the **rate of change of flux** ϕ_B will **induce an emf** given by:

$$\epsilon = - \frac{d\phi_B}{dt} = - Bl \frac{dx}{dt}$$

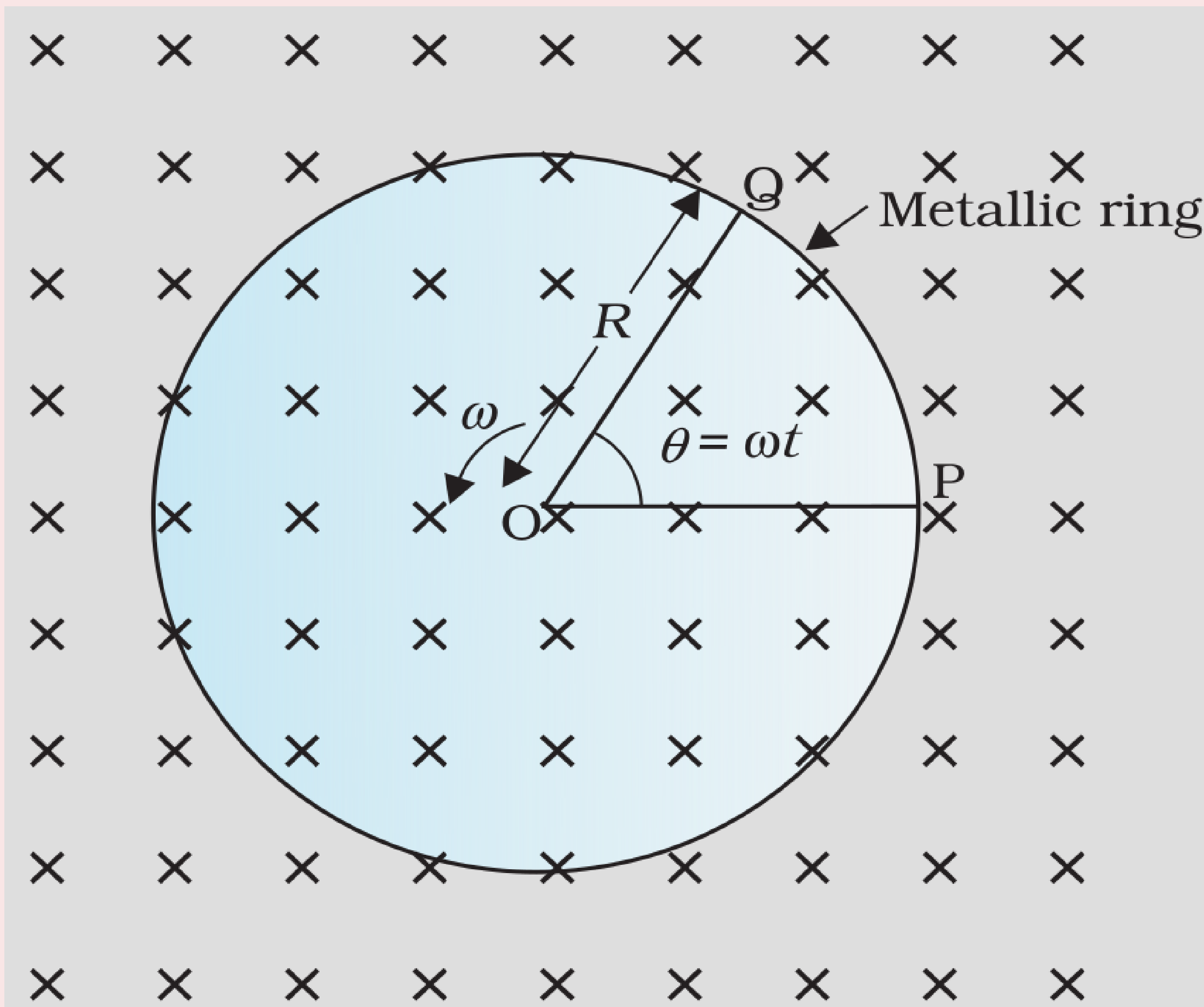
Motional Electromotive Force



$$\epsilon = Blv \quad \left(\because \frac{dx}{dt} = v \right)$$

- The induced emf Blv is called **motional emf**.

Example 6.6



A metallic rod of 1 m length is rotated with a frequency of 50 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring (Fig. 6.11). A constant and uniform magnetic field of 1 T parallel to the axis is present everywhere. What is the emf between the centre and the metallic ring?

Example 6.7

A wheel with 10 metallic spokes each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of earth's magnetic field H_E at a place. If $H_E = 0.4 \text{ G}$ at the place, what is the induced emf between the axle and the rim of the wheel? Note that $1 \text{ G} = 10^{-4} \text{ T}$.

Exercise 6.4

A rectangular wire loop of sides 8 cm and 2 cm with a small cut is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the loop. What is the emf developed across the cut if the velocity of the loop is 1 cm s^{-1} in a direction normal to the (a) longer side, (b) shorter side of the loop? For how long does the induced voltage last in each case?

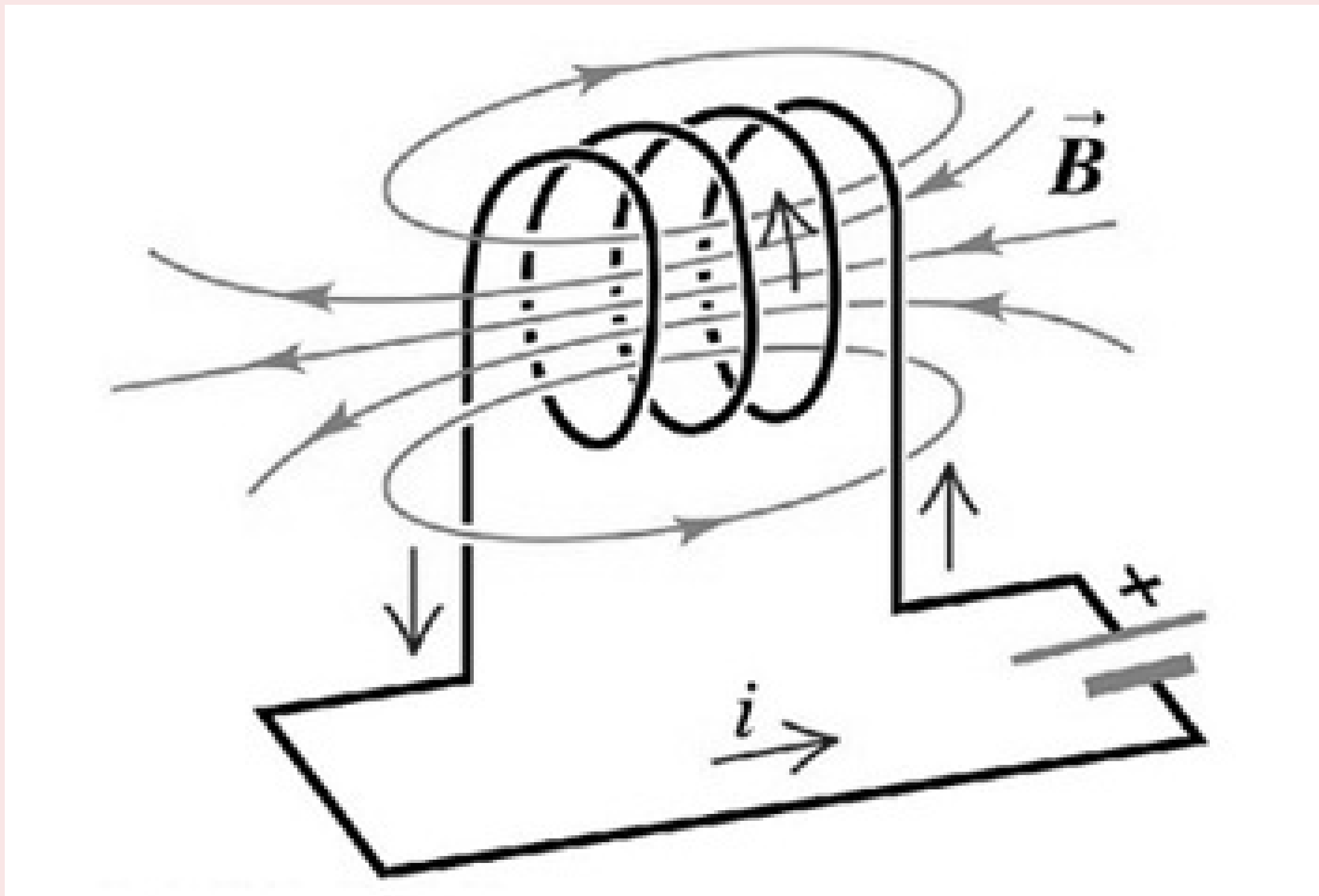
Exercise 6.5

A 1.0 m long metallic rod is rotated with an angular frequency of 400 rad s^{-1} about an axis normal to the rod passing through its one end. The other end of the rod is in contact with a circular metallic ring. A constant and uniform magnetic field of 0.5 T parallel to the axis exists everywhere. Calculate the emf developed between the centre and the ring.

INDUCTANCE

- In **Electromagnetic Induction**, we discussed how a **time-varying magnetic flux induces an emf** (current) in a circuit.
- The **reverse of this phenomenon** also occurs: A **varying current results in a varying magnetic field**. Due to this, the **magnetic flux varies** and an **electromotive force is induced** in the circuit.
- **Inductance** is the **tendency of an electrical conductor to oppose a change in the electric current** flowing through it.
- **Inductance** is the **property of a device that tells us how effectively it induces an emf in another device**.

Self-Inductance



- **An induced EMF appears in a single coil if the current in that coil is changing.** This phenomenon is known as **self-inductance**.
- The **current produces a magnetic field** which passes through every turn of the coil. If the **current is changing** with time, the **magnetic field will also change**.
- Consequently the **magnetic flux** passing through the coil **changes, resulting** in an **induced EMF**.
- This EMF is called the self induced EMF and it obeys Faraday's law of induction.

Self-Inductance

- For a given coil, the magnetic flux linked with it will be proportional the current.

$$\phi \propto I \quad \text{OR} \quad \phi = LI$$

where the constant of proportionality **L** is known as the **self-inductance** of the wire loop. **If the coil has N turns**, this equation becomes

$$N\phi = LI$$

- For a loop with N turns, using Faraday's law $\varepsilon = -N \frac{d\phi}{dt}$, so the induced emf is

$$\varepsilon = -L \frac{dI}{dt}$$

- **self-induced emf** always **opposes any change** (increase or decrease) of **current** in the coil.

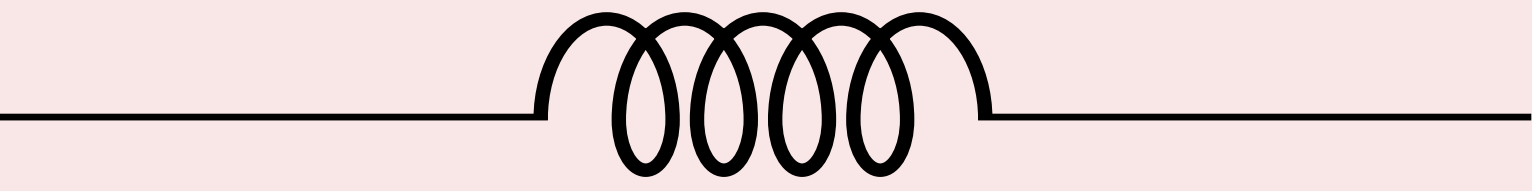
Self-Inductance

- **Definition** : Self-inductance is defined as the **magnetic flux** associated with the coil when **unit current** flows through it.

OR

- **Definition** : Self-inductance is defined as the **emf induced** in the coil when **the rate of change of current** through the coil is **unity**.
- The **self-inductance** of the coil **depends on** its **geometry** and on the **permeability** of the medium.
- The **self-induced emf** is **also called** the **back emf** as it opposes any change in the current in a circuit.
- **self-inductance** plays the role of **inertia**. It is the **electromagnetic analogue of mass** in mechanics.

Self-Inductance

- S.I. Unit of inductance is **henry** denoted by **H**
- Inductance is a **scalar quantity** having dimensions $[ML^2T^{-2}A^{-2}]$
- Symbol of inductor :  OR 

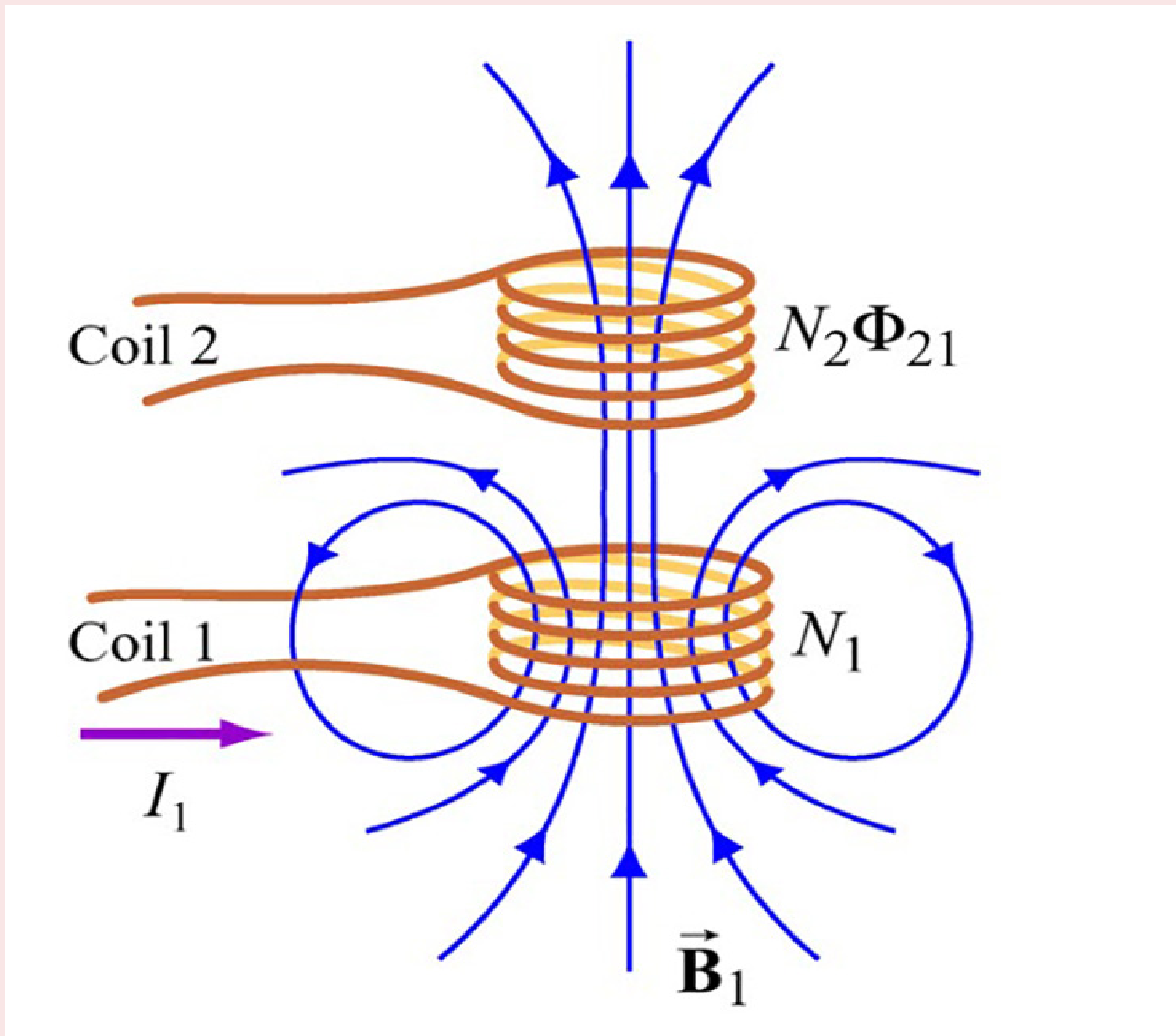
Self-Inductance of a long Solenoid

- Consider a long solenoid of **cross-sectional area** A and **length** l , having $\left(n = \frac{N}{l}\right)$ **turns per unit length**.
- The **magnetic field** due to a **current** I flowing in the solenoid is : $B = \mu_0 n I$
- The **total flux linked** with the **solenoid** is : $N\phi = (nl)(\mu_0 n I)A$

$$N\phi = \mu_0 n^2 l A I \quad \dots\dots(1)$$

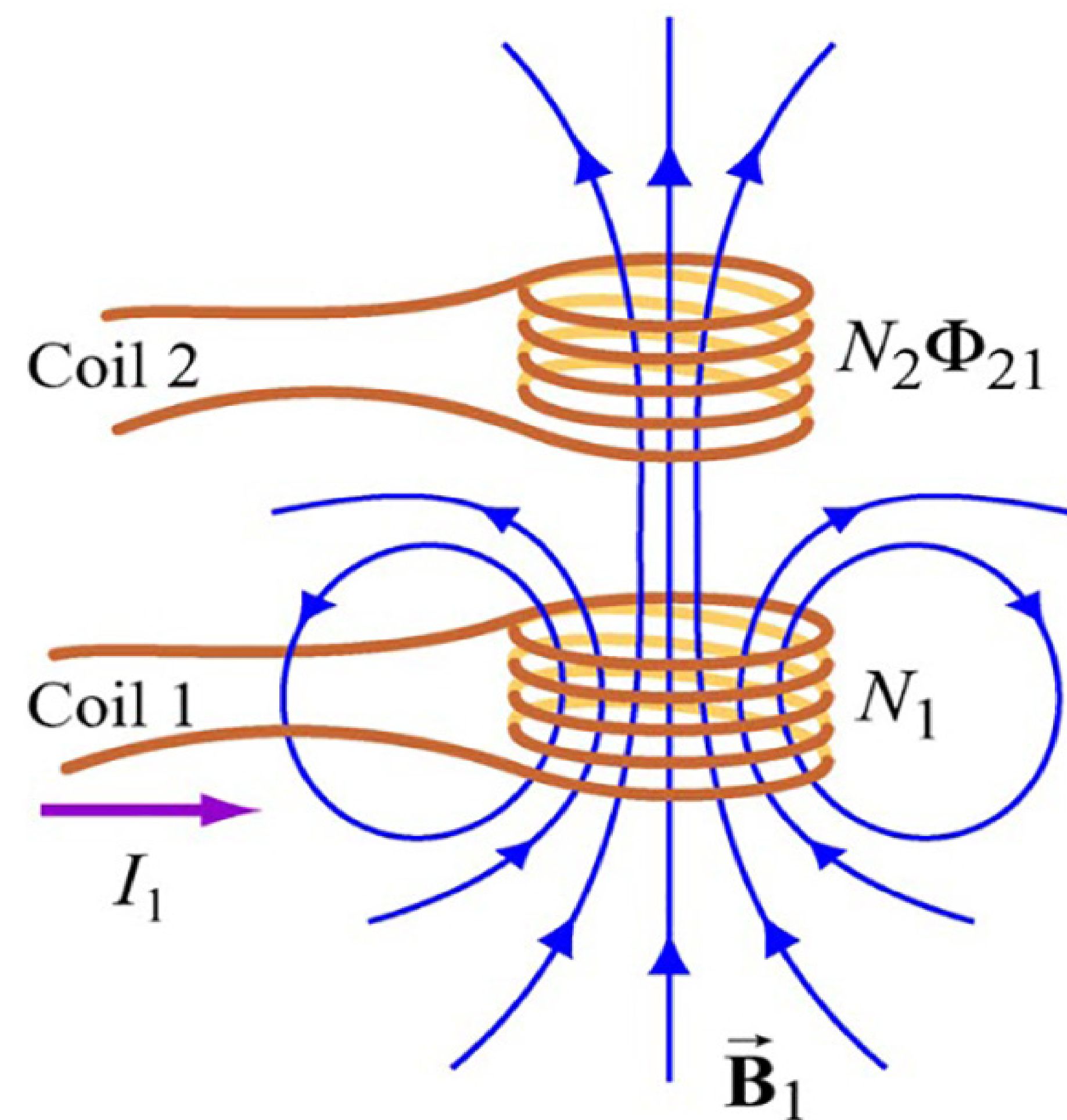
- Comparing above equation with $N\phi = LI$ we get :
- **Self inductance of solenoid** : $L = \mu_0 n^2 l A$

Mutual-Inductance



- **Mutual inductance** is the phenomenon in which a **change in current in one coil induces an EMF in another nearby coil** due to the shared magnetic field.
- When **current (I_1) in coil 1 changes it changes the magnetic field (B_1)** and therefore the **flux linked through the coil 2 ($N_2\phi_{21}$) also changes**.
- This **flux linked with the coil 2 is directly proportional to the current flowing in the coil 1**
$$N_2\phi_{21} \propto I_1$$
- Here, M_{21} is called **mutual inductance of second coil 2 w.r.t coil 1**.

Mutual-Inductance

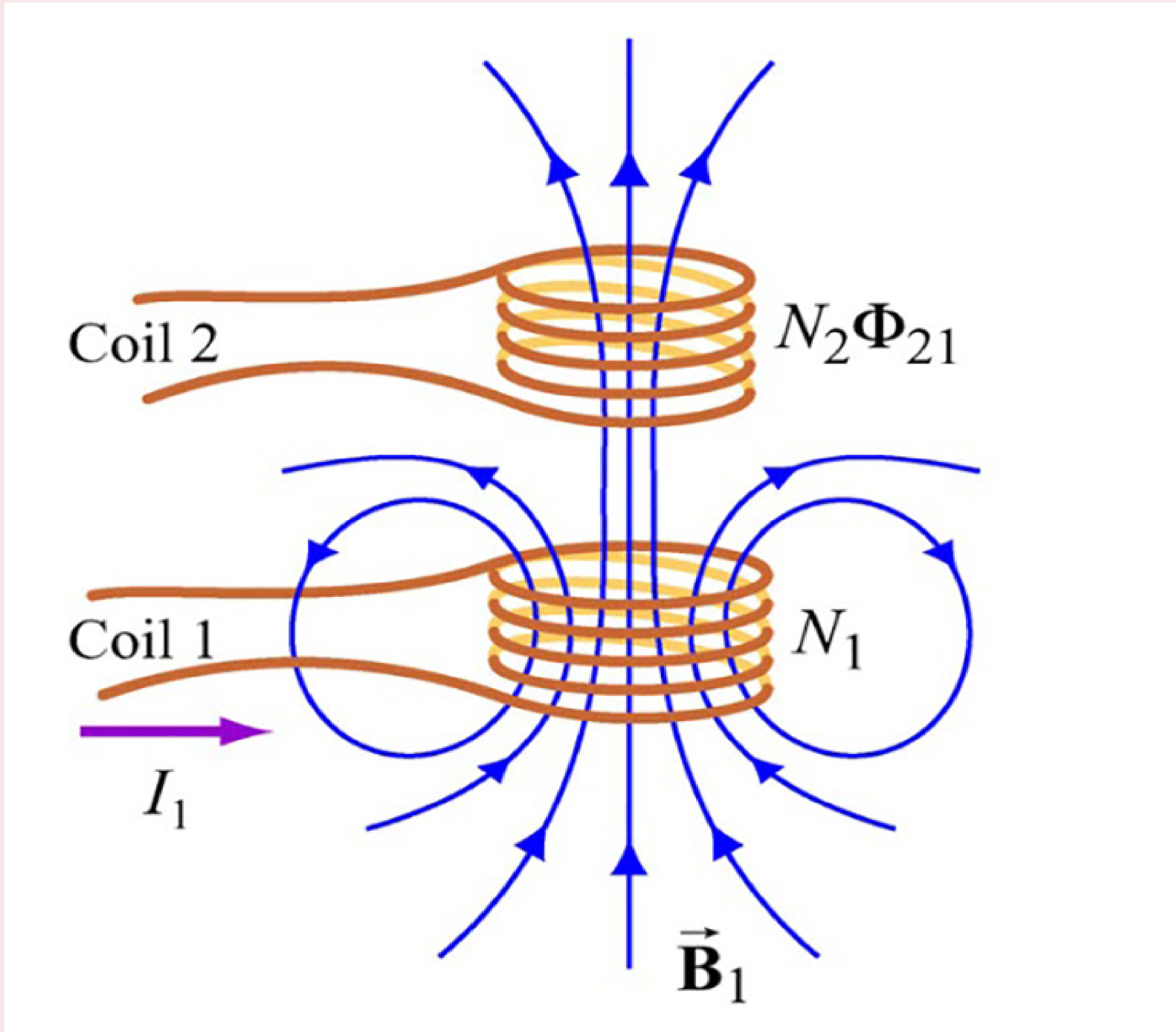


- The emf developed in either coil is found by combining Faraday's law and the definition of mutual inductance. Since $N_2\phi_{21}$ is the total flux through coil 2 due to I_1 , we obtain

$$\varepsilon_2 = -\frac{d}{dt}(N_2\phi_{21}) = -\frac{d}{dt}(M_{21}I_1)$$

$$\boxed{\varepsilon_2 = -M_{21}\frac{dI_1}{dt}}$$

Mutual-Inductance

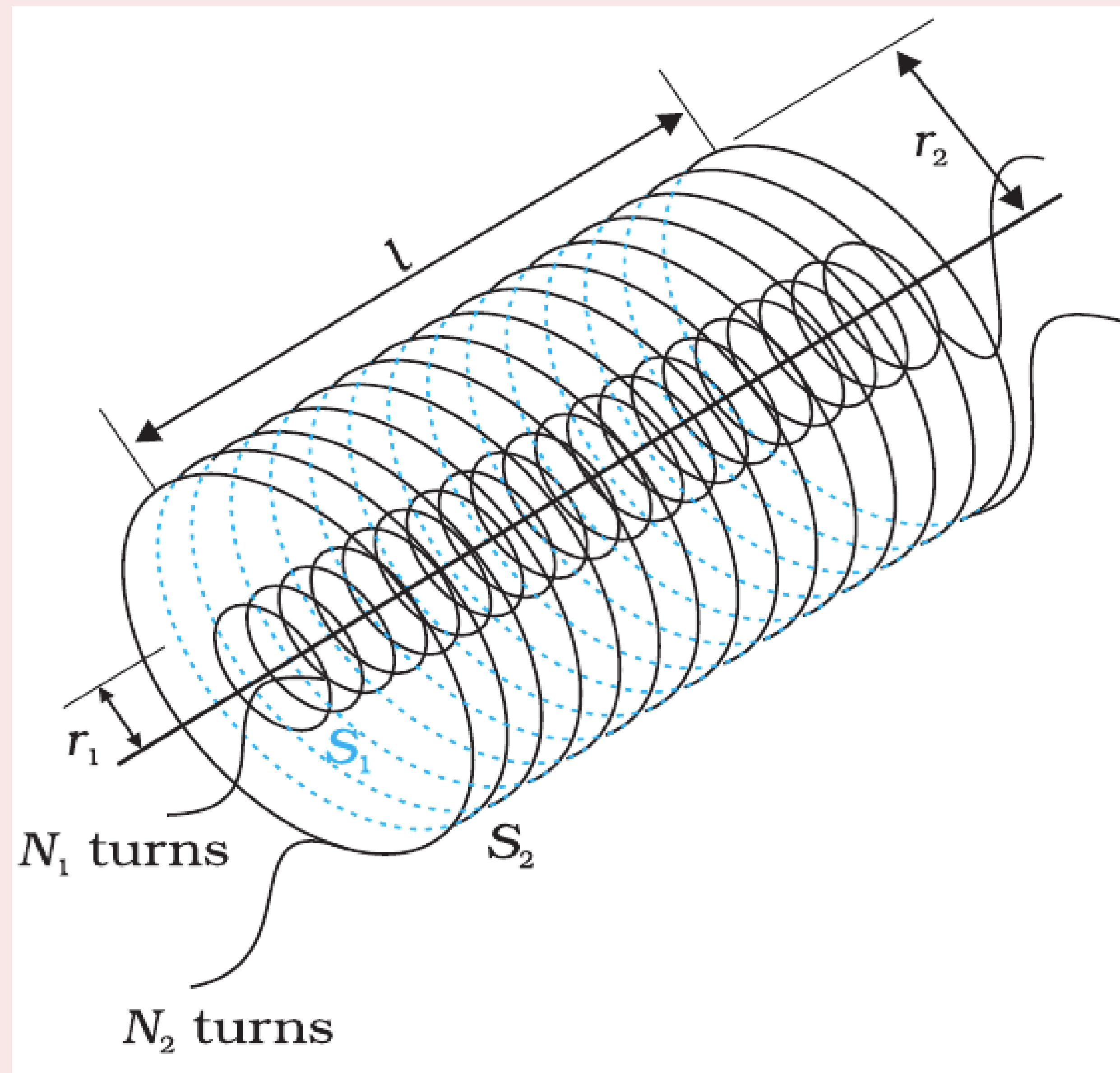


- **Definition :** Mutual inductance is the **magnetic flux associated with the secondary coil** due to the **flow of unit current** in the **primary coil**.

OR

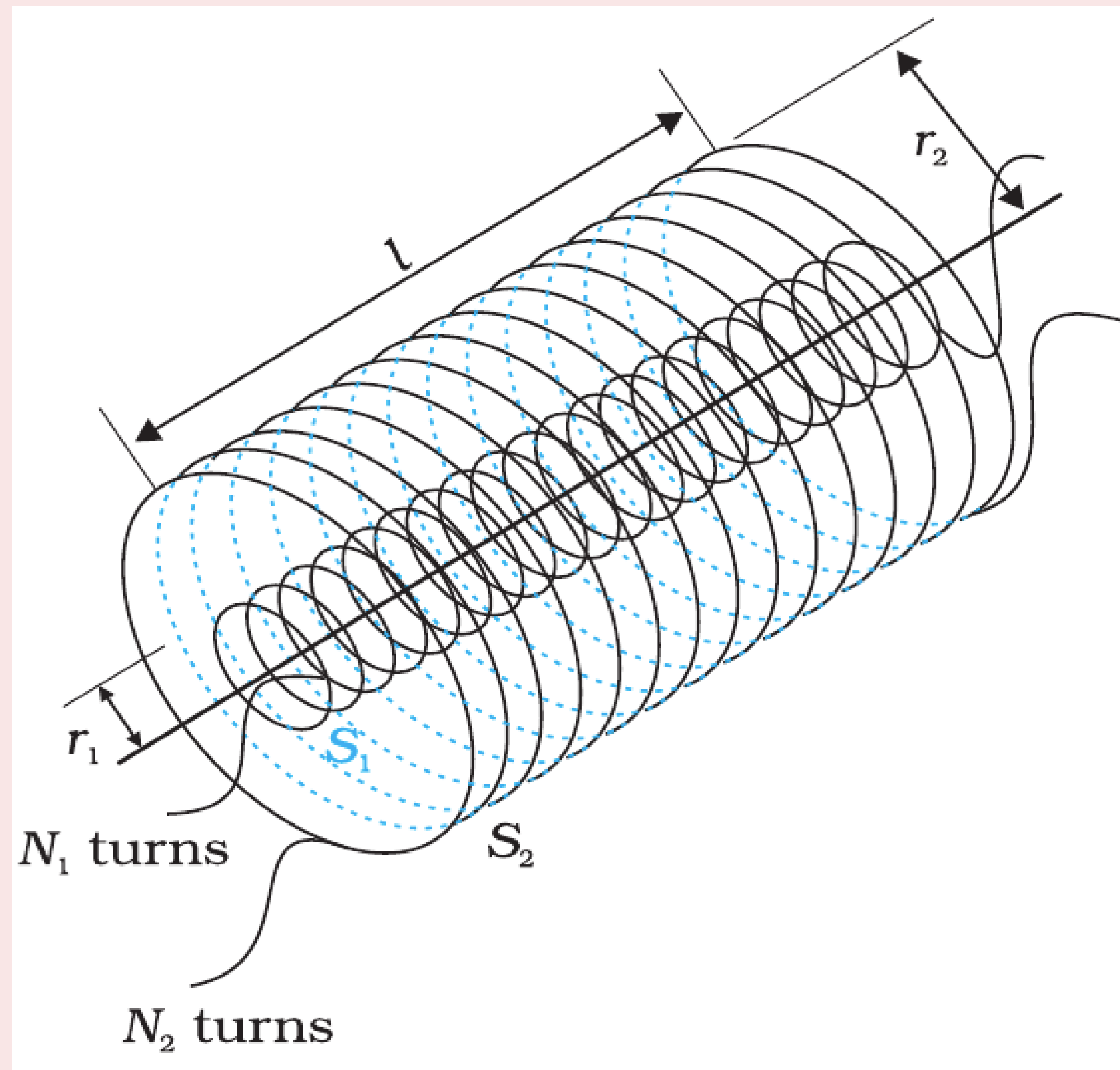
- **Definition :** Mutual inductance is the **emf induced in the secondary coil** when **rate of change of current** in the **primary coil is unity**.
- The **mutual-inductance** of pairs coils or solenoids etc., **depends on their geometry, separation** as well as their relative **orientation**, and on the **permeability** of the medium, but is **independent of the currents** in the coils.

Mutual-Inductance Two Long Co-axial Solenoids



- Consider **two long co-axial solenoids** S_1 and S_2 each of **length** l .
- N_1 and $N_2 \rightarrow$ **Total number of turns** of S_1 and S_2 .
- $n_1 = \frac{N_1}{l}$ and $n_2 = \frac{N_2}{l} \rightarrow$ **number of turns per unit length** of S_1 and S_2 .
- r_1 and $r_2 \rightarrow$ **Radius** of inner S_1 and outer S_2 solenoids.
- When **current** I_2 flow through the **outer coil** S_2 the **magnetic field** inside it is $B_2 = \mu_0 n_2 I_2$
- Total flux linkage with solenoid S_1 is $N_1 \phi_{12} = N_1 B_2 A_1$

Mutual-Inductance Two Long Co-axial Solenoids

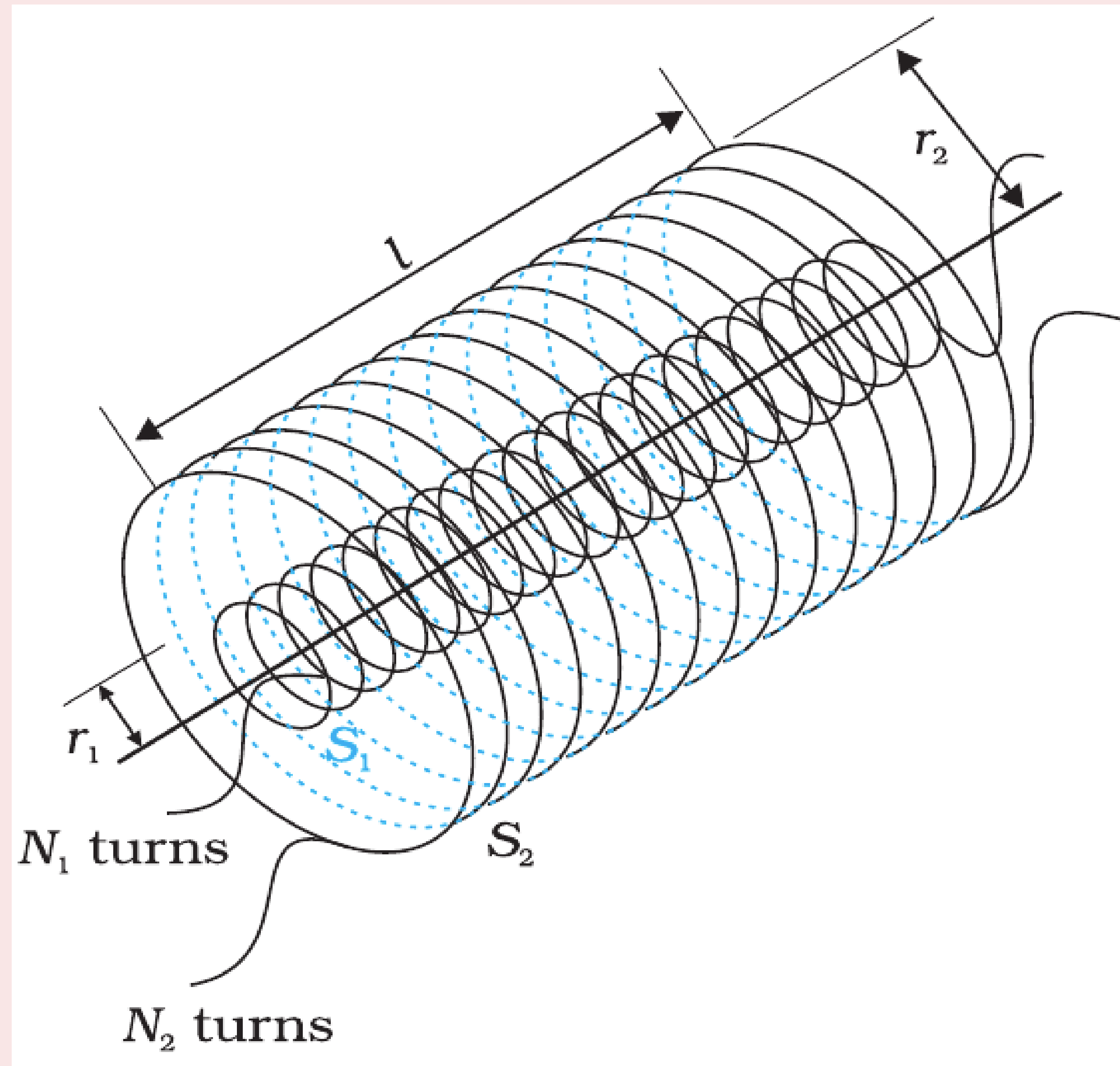


- $N_1 \phi_{12} = \mu_0 n_1 n_2 l \pi r_1^2 I_2$
- Comparing with $N_1 \phi_{12} = M_{12} I_2$ we get
- Mutual inductance of solenoid S_1 with respect to S_2

$$M_{12} = \mu_0 n_1 n_2 l \pi r_1^2 \quad \dots\dots(1)$$

- Now consider the **reverse case**. A **current I_1** is passed through the **solenoid S_1**
- $N_2 \phi_{21} = \mu_0 n_1 n_2 l \pi r_1^2 I_1$
- Comparing with $N_2 \phi_{21} = M_{21} I_1$ we get

Mutual-Inductance Two Long Co-axial Solenoids



- Mutual inductance of solenoid S_2 with respect to S_1

$$M_{21} = \mu_0 n_1 n_2 l \pi r_1^2 \dots\dots(2)$$

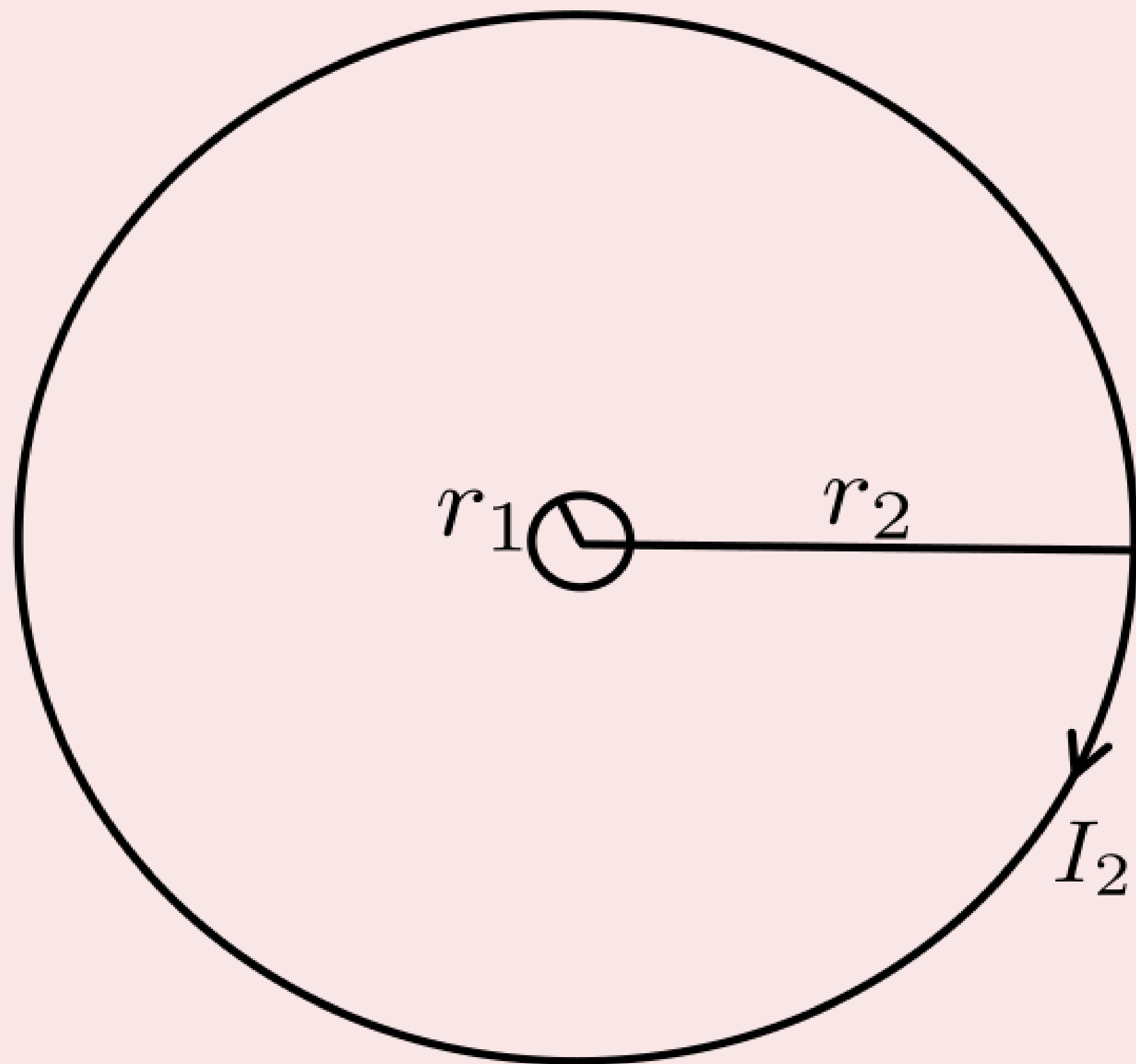
- From equation (1) and (2) We get,

$$M_{12} = M_{21} = M$$

- If a medium of relative permeability μ_r had been present, the mutual inductance would be

$$M = \mu_r \mu_0 n_1 n_2 l \pi r^2$$

Mutual-Inductance Two concentric circular coils ($r_1 \ll r_2$)



- Let a **current I_2** flow through the **outer circular coil** .
- The **magnetic field** at the **centre** of the **coil 2** is-

$$B_2 = \frac{\mu_0 I_2}{2r_2}$$

- Flux passing through inner coil is -

$$\phi_{12} = B_2 A_1$$

- Comparing with $\phi_{12} = M_{12} I_2$ we get

- $$M_{12} = \frac{\mu_0 \pi r_1^2}{2r_2}$$

Energy Stored in an Inductor

- In an inductor **work needs to be done against the back emf ε** in establishing the current. **This work done is stored as magnetic potential energy.**

$$\begin{aligned}P &= \frac{dW}{dt} = |\varepsilon|I \\ \frac{dW}{dt} &= L \frac{dI}{dt} I \\ dW &= LI dI \\ \int dW &= L \int I dI \\ W &= \frac{1}{2} LI^2\end{aligned}$$

Example 6.10 (a) Obtain the expression for the magnetic energy stored in a solenoid in terms of magnetic field B , area A and length l of the solenoid. (b) How does this magnetic energy compare with the electrostatic energy stored in a capacitor?

Solution

(a) From Eq. (6.19), the magnetic energy is

$$\begin{aligned} U_B &= \frac{1}{2} LI^2 \\ &= \frac{1}{2} L \left(\frac{B}{\mu_0 n} \right)^2 && \text{(since } B = \mu_0 nI, \text{ for a solenoid)} \\ &= \frac{1}{2} (\mu_0 n^2 Al) \left(\frac{B}{\mu_0 n} \right)^2 && \text{[from Eq. (6.17)]} \\ &= \frac{1}{2\mu_0} B^2 Al \end{aligned}$$

(b) The magnetic energy per unit volume is,

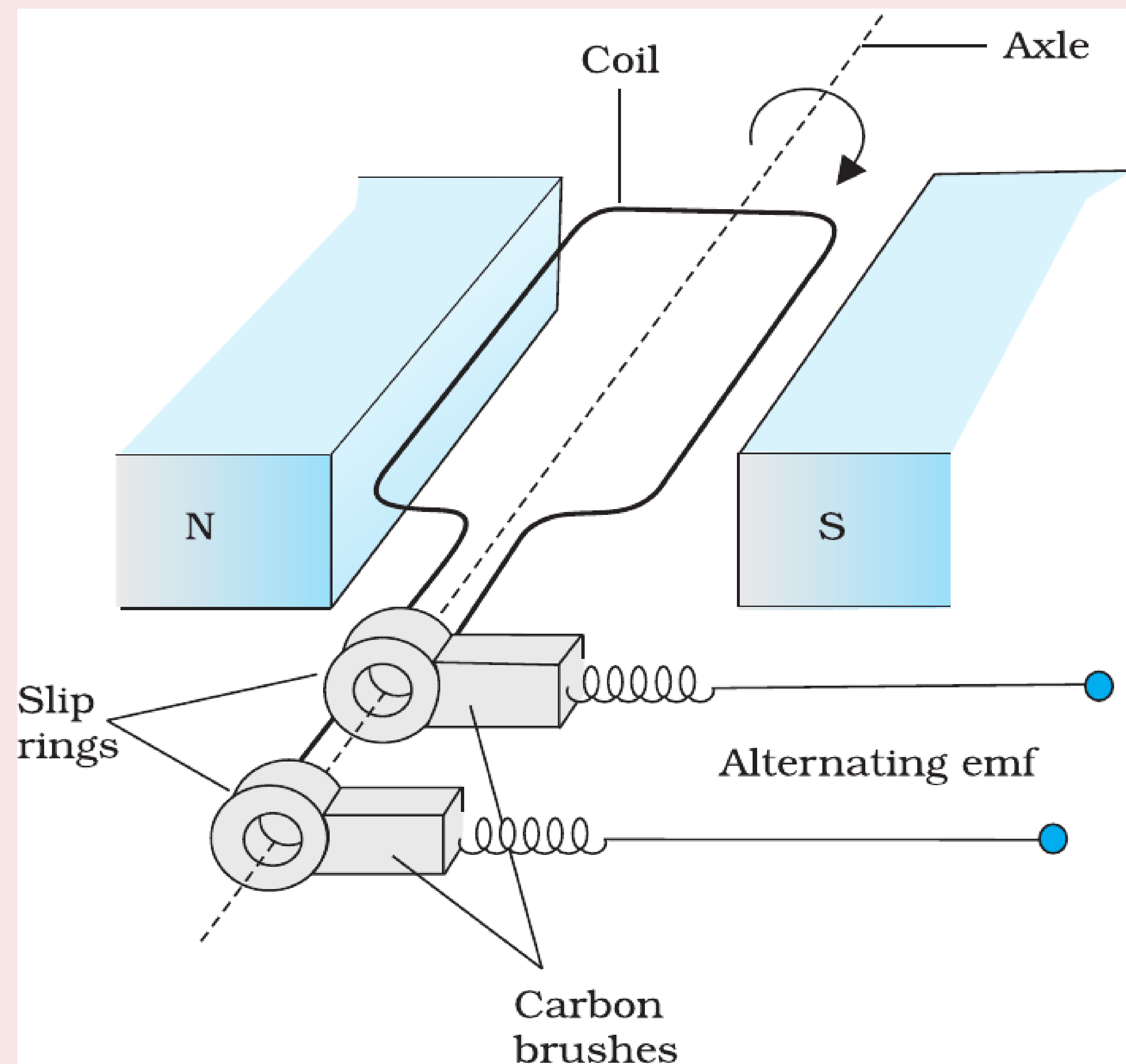
$$\begin{aligned} u_B &= \frac{U_B}{V} && \text{(where } V \text{ is volume that contains flux)} \\ &= \frac{U_B}{Al} \\ &= \frac{B^2}{2\mu_0} \end{aligned} \tag{6.20}$$

We have already obtained the relation for the electrostatic energy stored per unit volume in a parallel plate capacitor (refer to Chapter 2, Eq. 2.77),

$$u_E = \frac{1}{2} \epsilon_0 E^2 \tag{2.77}$$

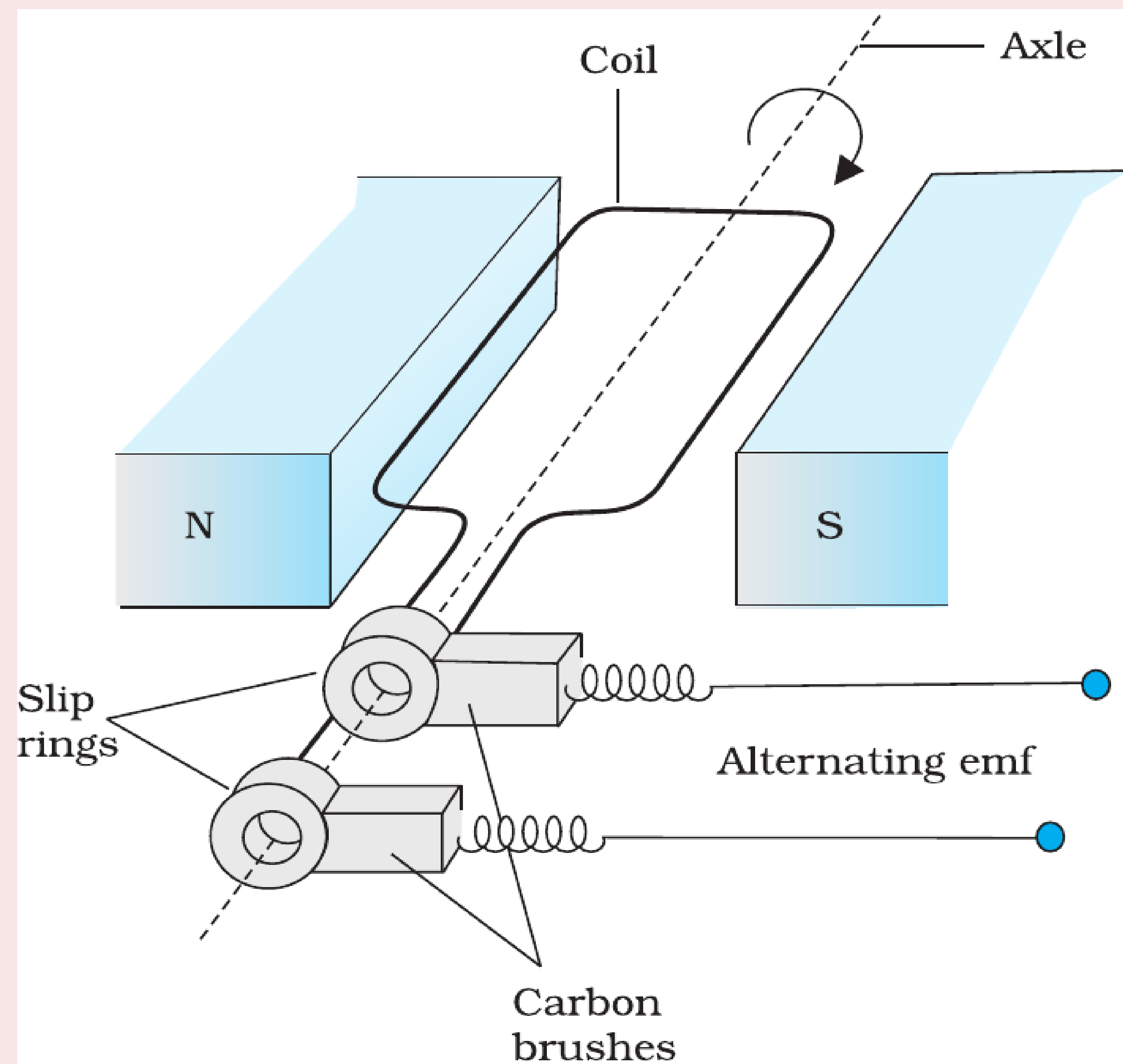
In both the cases energy is proportional to the square of the field strength. Equations (6.20) and (2.77) have been derived for special cases: a solenoid and a parallel plate capacitor, respectively. But they are general and valid for any region of space in which a magnetic field or/and an electric field exist.

AC GENERATOR



- An ac generator **converts mechanical energy into electrical energy.**
- **Working Principle :** Its working is based on electromagnetic induction.
- **Construction :** It consists of a **magnet**, an **armature (coil)** with **slip rings** and **brushes.**
 - ① **Magnet :** It may be a **permanent magnet** or an **electromagnet** and provide a strong uniform magnetic field.
 - ② **Armature :** It is a **coil** generally **wound over a soft-iron core.** The **core increases the magnetic field** due to its magnetization.

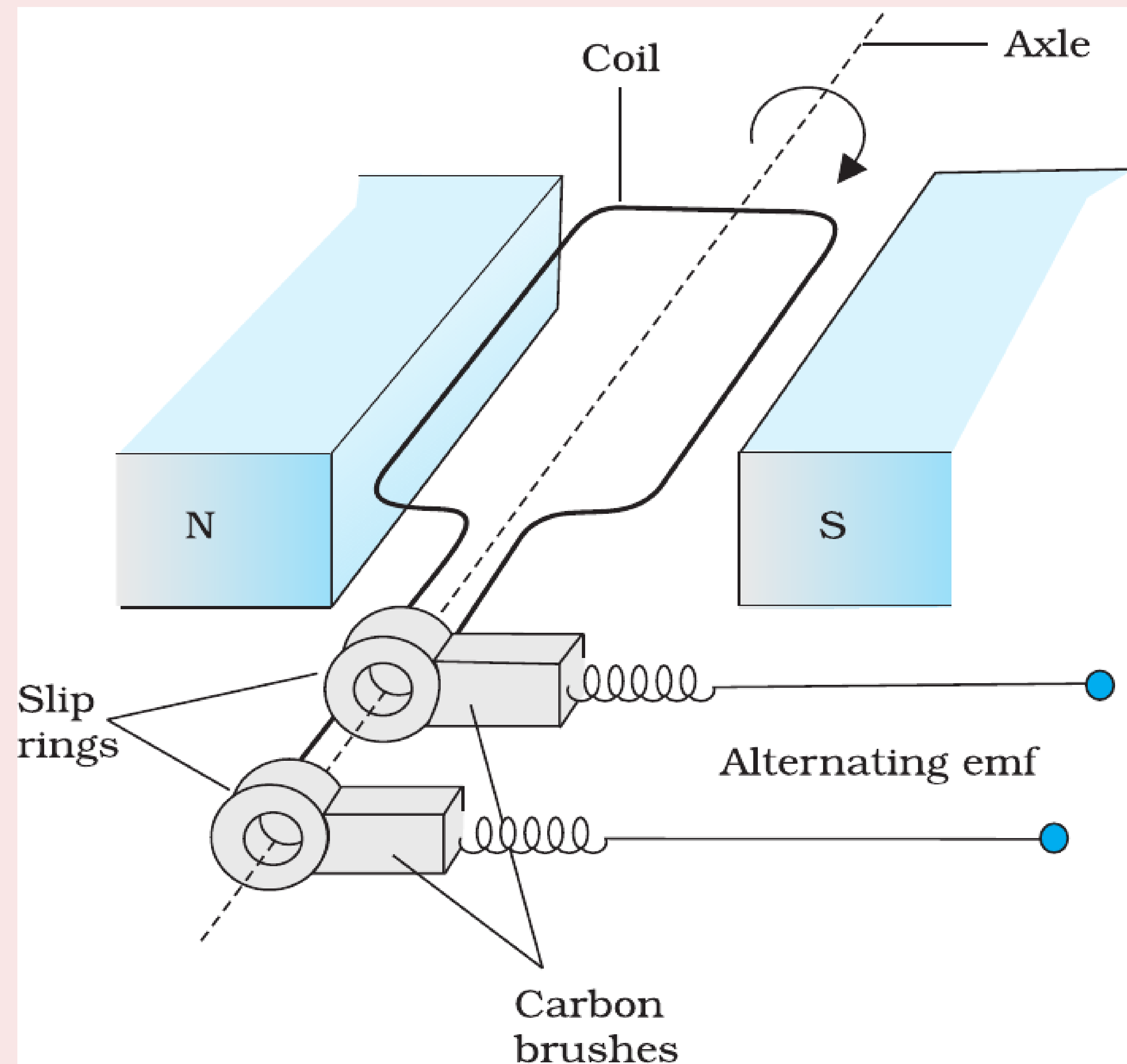
AC GENERATOR



- **3 Slip Rings :** The **two ends of the coil are connected to two slip rings** . The coil together with the rings can rotate in the magnetic field.
- **4 Carbon Brushes :** **Two graphite brushes permanently touch the slip rings**. As the armature rotates, the slip rings slip against the brushes so that the **contact is maintained all the time**. These brushes are connected to the external circuit.

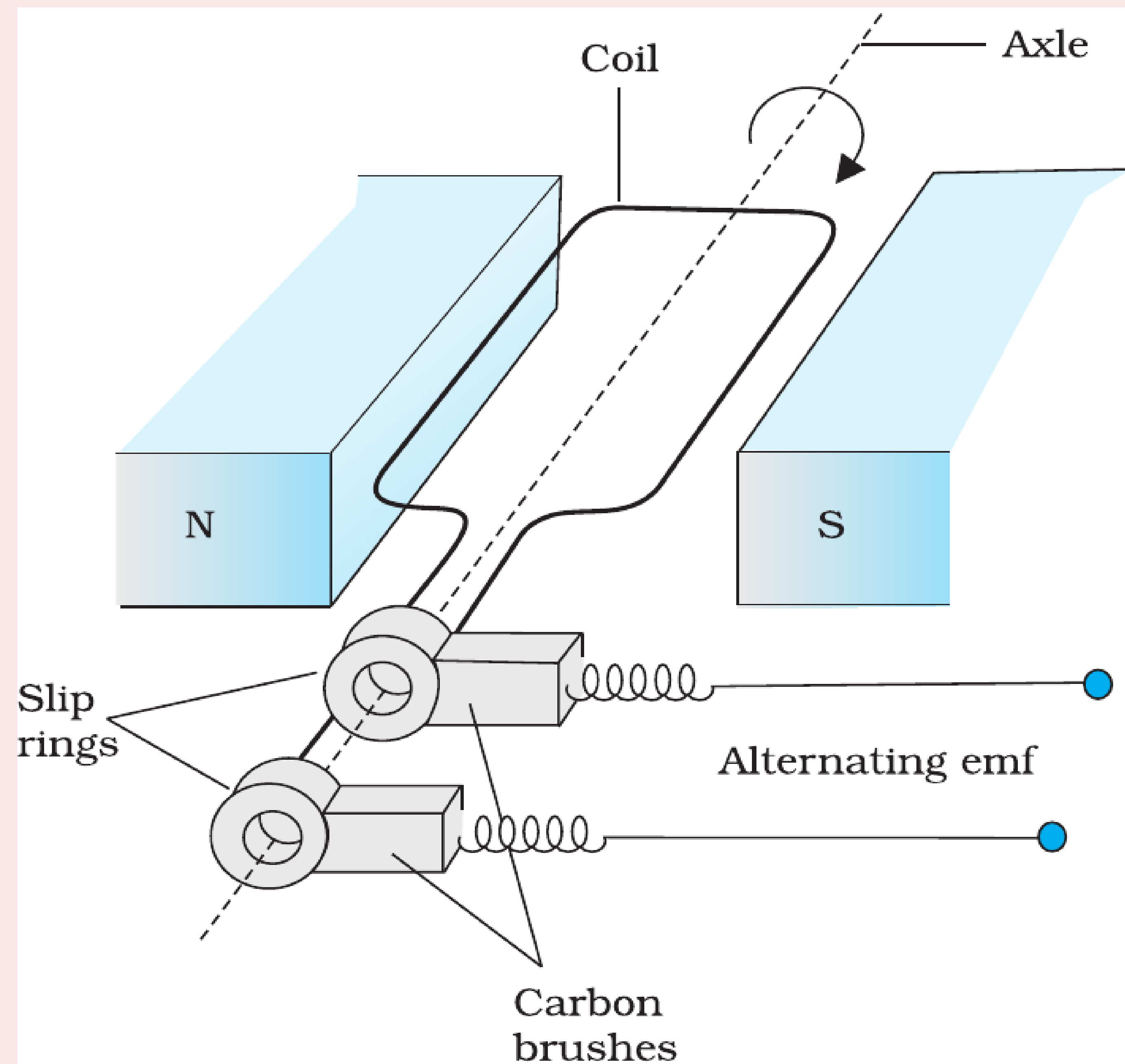
AC GENERATOR

Theory and Working



- When the coil is rotated with a constant angular speed ω , area vector $\vec{A}$ of the coil at any instant t is $\theta = \omega t$ (assuming $\theta = 0^\circ$ at $t = 0$).
- Total magnetic flux linked with coil at any time t is :
$$\phi = B A \cos \theta$$
- From Faraday's law, the induced emf for the rotating coil of N turns is
$$\varepsilon = - N \frac{d\phi}{dt}$$

AC GENERATOR



- $\varepsilon = N B A \omega \sin(\omega t)$
- The **maximum** magnitude of the **emf**, known as **peak emf**, is $\varepsilon_0 = N B A \omega$. which occurs when $\sin \omega t = \pm 1$. then, $\varepsilon = \varepsilon_0 \sin(\omega t)$
- The **direction of the current changes periodically** and therefore the current is called **alternating current (ac)**.
- Since $\omega = 2\pi\nu$, Therefore, $\varepsilon = \varepsilon_0 \sin(2\pi\nu t)$ where ν is the **frequency of revolution of the generator's coil**.

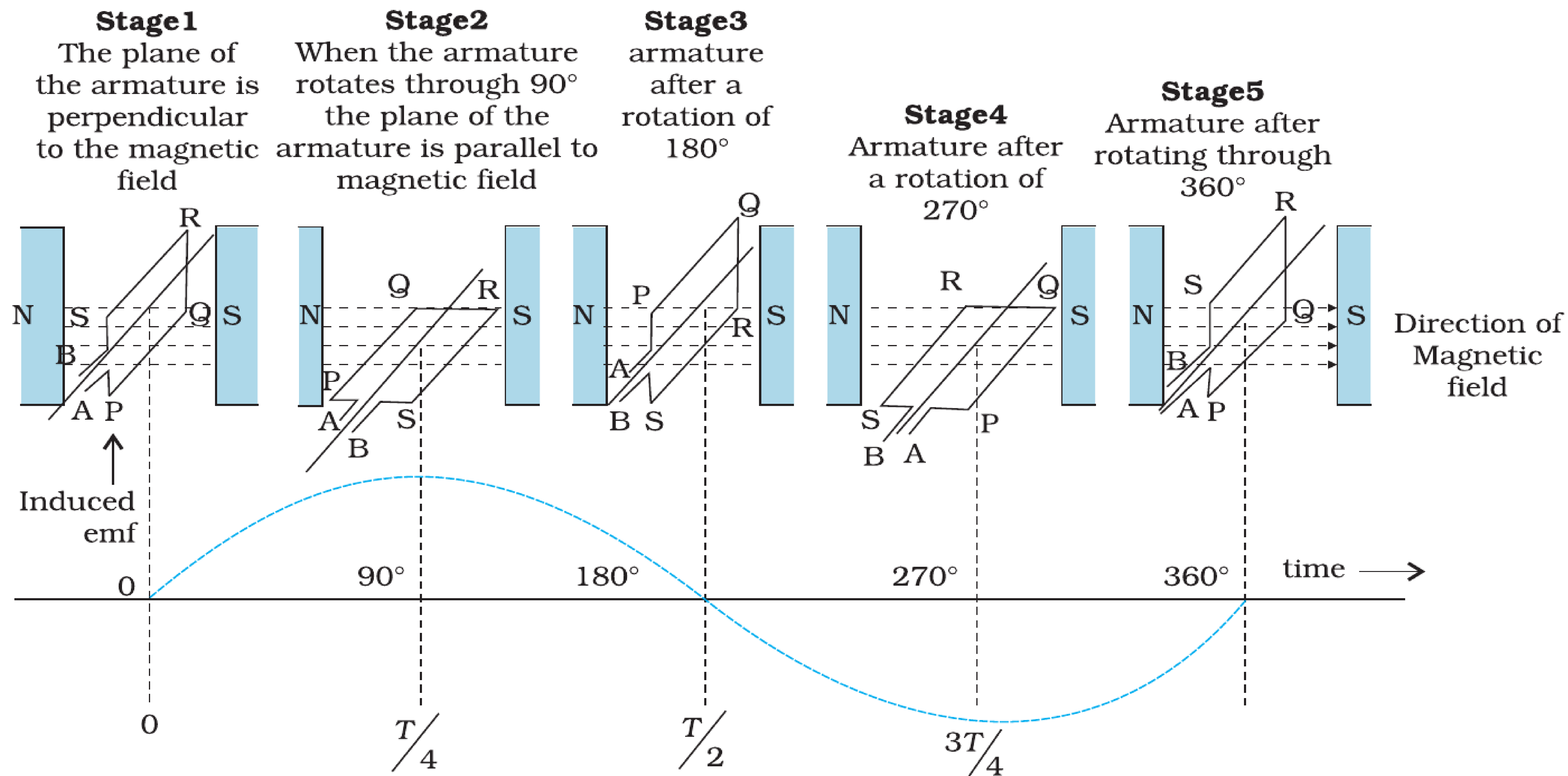


FIGURE 6.17 An alternating emf is generated by a loop of wire rotating in a magnetic field.

Example 6.11 Kamla peddles a stationary bicycle. The pedals of the bicycle are attached to a 100 turn coil of area 0.10 m^2 . The coil rotates at half a revolution per second and it is placed in a uniform magnetic field of 0.01 T perpendicular to the axis of rotation of the coil. What is the maximum voltage generated in the coil?

Solution Here $\nu = 0.5 \text{ Hz}$; $N = 100$, $A = 0.1 \text{ m}^2$ and $B = 0.01 \text{ T}$. Employing Eq. (6.21)

$$\begin{aligned}\varepsilon_0 &= NBA (2 \pi \nu) \\ &= 100 \times 0.01 \times 0.1 \times 2 \times 3.14 \times 0.5 \\ &= 0.314 \text{ V}\end{aligned}$$

The maximum voltage is 0.314 V .

We urge you to explore such alternative possibilities for power generation.